Qualcomm Technologies, Inc.

Hardware Overview

WRD6100

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80-83950-130 Rev. AA

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Revision History

Revision	Date	Description
AA	September 2025	Initial release.

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Section 1

Introduction

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Device Scope, Assumptions, and Requirements

The Watch Form Factor Reference Platform is a prototype designed to demonstrate Qualcomm's SoC capabilities in smartwatches, featuring optimized power, performance, and thermal management, while enabling customers to accelerate design cycles, validate integrations.

- Product vision for the Wearable Reference Design (WRD)
  - Design and build the device first to help understand and address key customer challenges
    - Implement the SW6100 chipset in a watch form factor representative platform that supports both Sub-6 5G and mm Wave technologies.
    - Solve key technology challenges in the industrial/mechanical/electrical design and integration of 4G/5G antennas, Qualcomm® RF Front End (RFFE) architecture, and other key chipset features.
    - Demonstrate Qualcomm Technologies, Inc. (QTI) value-add, and advanced 5G feature sets and key performance indicators (KPIs) including power, performance, and thermal.
- Product requirements
  - WRD6100 (SKU1/SKU2) has the following priorities:
    - ID form factor, display, and battery size reflecting OEM requirements.
    - PCB design reflecting OEM constraints in area, shape, layers, component placement, and design rules.
    - Modem, CPU/GPU, cameras, haptics, audio, connectivity features, and KPIs to reflect OEM requirements.
    - Debug capabilities are prioritized by PCB size constraints.
    - Availability to enable internal software development, industry demos, and use in various test settings.

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NOTE: This device does not attempt to fully feature or exercise every capability of the chipset. Instead, its goal is to showcase key 4G/5G technology in a form factor representative of a commercial product while enabling a subset of possible features and bands that would be expected in a commercial design. This device is based on a 1.39-inch display form factor and is designed to perform at limits associated with this overall size.

Section 2

SW6100 Pin Mapping- SKU1 vs SKU2

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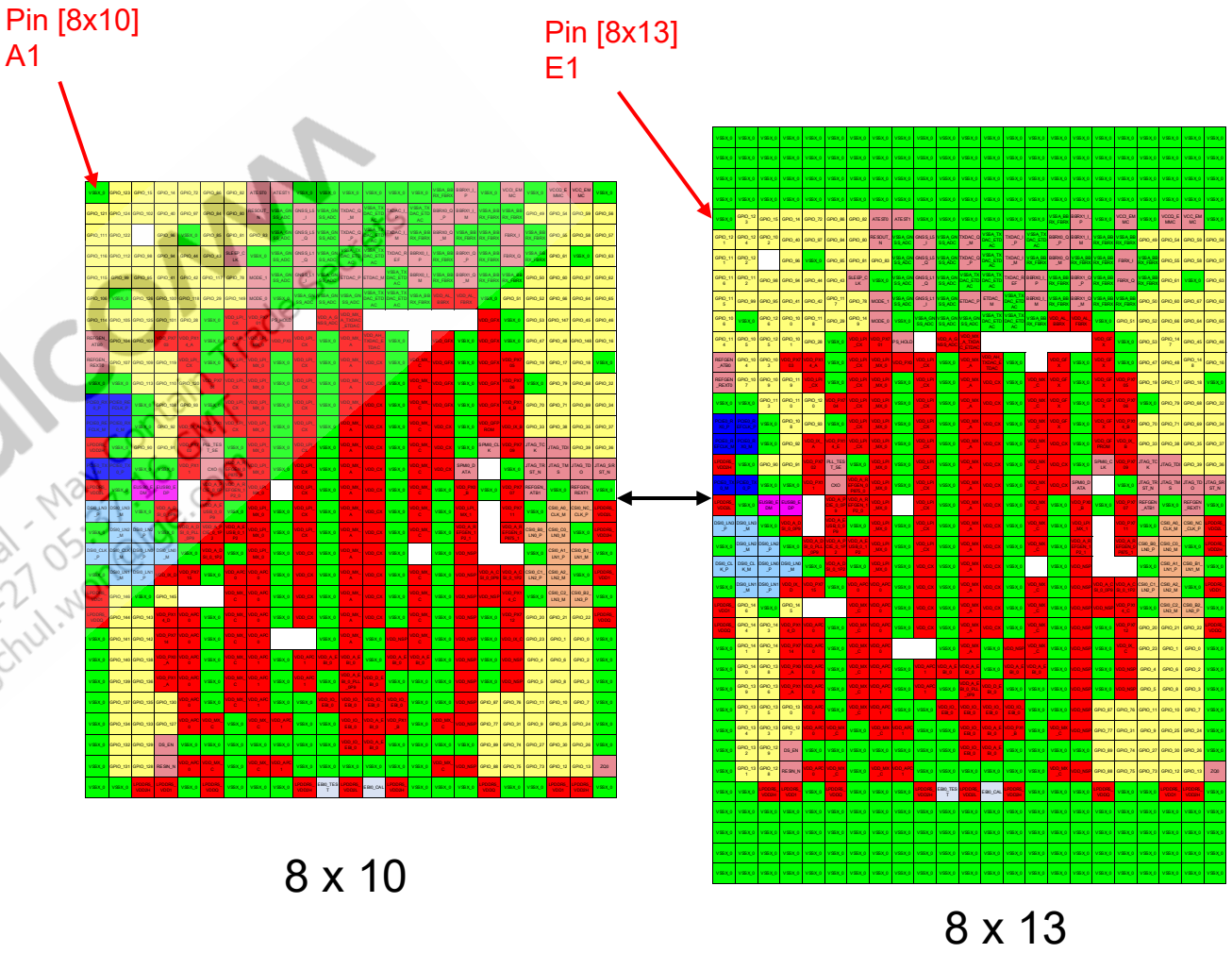
SW6100 Pin Mapping- SKU1 vs SKU2

- 8x13 is Superset of 8x10
- 8x10 can be installed on 8x13 footprint
- Pin [8x10] A1 maps to Pin[8x13] E1

SW6100 Variant	Bottom Pin Map
SW6100-0/1	8x10 RDL/MEP
SW6100-2	8x13 RDL

WRD6100 SKUs			
	BB	PMIC	RF
SKU1	SW6100-0/1	PMW6100	ET
SKU2	SW6100-0/1/2	PW6100-0/1	APT



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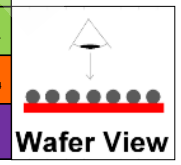
PW6100-0 and PW6100-1 Pin Mapping

- **PW6100-0 and PW6100-1 are Pin to Pin compatible**
- **PW6100-0 Pin(16) Maps to PW6100-1 (A1)**

PW6100-0 (16)

PW6100-0 : Bottom View

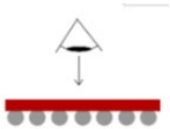
320 VREG_XO	304 VDD_XO_RF	288 VDD_XO_RF	272 USB_DP	256 USB_DM	240 EUSB_EDM	224 VDD_BOB	208 VSW_BCK_BOB	192 VSW_BST_BOB	176 VREG_BOB	160 OVP_DRV	144 OVP_SNS	128 VBATT_PWR	112 VPH_PWR	96 PGND_CHG	80 VSW_CHG	64 CHG_MID	48 CHG_IN	32 CHG_IN	16 CHG_IN
319 XTAL_OUT	303 VREG_XO	287 VREG_RF	271 VDD_3V_EUSB	255 GND_EUSB	239 EUSB_EDP	223 VDD_BOB	207 VSW_BCK_BOB	191 PGND_BOB	175 VREG_BOB	159 CHG_BYP_EN	143 SBU1	127 VBATT_PWR	111 VPH_PWR	95 PGND_CHG	79 VSW_CHG	63 CHG_MID	47 CHG_IN	31 CHG_IN	15 CHG_IN
318 XTAL_IN	302 XTAL_OUT	286 GND_XO	270 VREG_1P6_EUSB	254 VDD_1P6_EUSB	238 VDD_1P2_EUSB	222 VDD_BOB	206 VDD_PDPHY	190 VREG_BOB_SNS	174 VREG_BOB	158 CC1_ID	142 SBU2	126 VBATT_SNS_M	110 VPH_PWR	94 BOOT_CAP	78 VARB_CHG	62 CHG_MID	46 CHG_MID	30 CMN_GND	14 CMN_GND
317 KPD_PWR_N	301 CMN_GND	285 CLK1_RF	269 CLK4_LN	253 CLK5_LN	237 VREF_IX_1	221 VREF_IX_2	205 AMUX_3	189 ADC_GND	173 XOADC_GND	157 CC2	141 CMN_GND	125 VBATT_SNS_P	109 VPH_PWR	93 CMN_GND	77 REF_GND_CHG	61 VREG_CHG_BST	45 VREG_CHG_BST	29 CMN_GND	13 CMN_GND
316 VREG_CBST	300 VREG_CBST_SNS	284 CLK3_RF	268 CLK2_RF	252 TEST_EN_VPP	236 VREF_TXDAC	220 AMUX_4	204 AMUX_2	188 RID	172 BATT_ID	156 XO_THERM	140 CMN_GND	124 CMN_GND	108 CMN_GND	92 CMN_GND	76 CMN_GND	60 CMN_GND	44 RMT_GND_S1	28 VDD_S1	12 VDD_S1
315 VSW_CBST	299 PGND_CBST	283 CMN_GND	267 CMN_GND	251 VDD_L23	235 VDD_L22	219 VDD_L20	203 VREG_L19	187 AMUX_1	171 BATT_THERM	155 REF_BYP	139 CMN_GND	123 VREG_1P8_SYS	107 VREG_1P2_SYS	91 VREG_L17	75 VREG_L11	59 CMN_GND	43 VREG_S1	27 VSW_S1	11 VSW_S1
314 VDD_CBST	298 CMN_GND	282 CMN_GND	266 SLEEP_CLK_IN	250 VREG_L23	234 VREG_L22	218 VREG_L20	202 VDD_L19_L21	186 VREG_L21	170 CMN_GND	154 REF_GND	138 XVDD_BYP	122 VIN_1P8_SYS	106 VDD_L12_L13	90 VDD_L7	74 VDD_L10_L11	58 VREG_L10	42 CMN_GND	26 GND_S1	10 GND_S1
313 GND_S11	297 GND_S11	281 CMN_GND	265 PM_SLEEP_CLK3	249 VREG_L26	233 VDD_L7_L26	217 VREG_L7	201 VREG_L30	185 VDD_L30	169 PM_I2C_CLK	153 CMN_GND	137 CMN_GND	121 VREG_L12	105 VREG_L13	89 VDD_L19_L16	73 VREG_L14	57 VDD_L14	41 CMN_GND	25 GND_S2	9 GND_S2
312 VSW_S11	296 VSW_S11	280 VREG_S11	264 PM_SLEEP_CLK2	248 VREG_L6	232 VDD_L6	216 VREG_L29	200 VDD_L29	184 VPH_PWR_INFRA_1	168 PM_I2C_DATA	152 PM_SPMI_CLK	136 PM_SPMI_DATA	120 VREG_L28	104 VREG_L15	88 VREG_L24	72 VREG_L16	56 CMN_GND	40 VREG_S2	24 VSW_S2	8 VSW_S2
311 VDD_S11	295 VDD_S11	279 RMT_GND_S11	263 PM_SLEEP_CLK1	247 VREG_L27	231 VDD_L8_L27	215 VDD_L9	199 VREG_L3	183 CMN_GND	167 CMN_GND	151 PM_GPIO_16	135 PM_GPIO_18	119 VDD_L18	103 VDD_L28	87 VDD_L24	71 VDD_L25	55 PM_GPIO_17	39 RMT_GND_S2	23 VDD_S2	7 VDD_S2
310 VDD_S10	294 VDD_S10	278 RMT_GND_S10	262 PMIC_RB	246 VREG_L8	230 VDD_L1	214 VDD_L4_L5	198 VREG_L4	182 CMN_GND	166 CMN_GND	150 PM_GPIO_14	134 PM_GPIO_15	118 VREG_L18	102 CMN_GND	86 VREG_L25	70 PM_GPIO_01	54 PM_GPIO_04	38 RMT_GND_S3	22 VDD_S3	6 VDD_S3
309 VSW_S10	293 VSW_S10	277 VREG_S10	261 PM_PON_RESET_N	245 VREG_L1	229 VDD_L3	213 VDD_L2	197 VREG_L5	181 CMN_GND	165 CMN_GND	149 PM_GPIO_12	133 PM_GPIO_13	117 CMN_GND	101 PM_GPIO_08	85 CMN_GND	69 PM_GPIO_02	53 PM_GPIO_03	37 VREG_S3	21 VSW_S3	5 VSW_S3
308 GND_S10	292 GND_S10	276 VREG_S9	260 RMT_GND_S9	244 VREG_L3_SNS	228 VREG_L3	212 VREG_L2	196 VREG_L2_SNS	180 CMN_GND	164 CMN_GND	148 PM_GPIO_11	132 PM_GPIO_10	116 PM_GPIO_09	100 PM_GPIO_07	84 PM_GPIO_06	68 PM_GPIO_05	52 RMT_GND_S4	36 VREG_S4	20 GND_S3	4 GND_S3
307 GND_S9	291 GND_S9	275 VSW_S9	259 VDD_HS_DRV_1	243 FAULT_N	227 RMT_GND_S8	211 VREG_S8	195 CMN_GND	179 VREG_S7	163 RMT_GND_S7	147 RMT_GND_S6	131 VREG_S6	115 VPH_PWR_INFRA_2	99 VREG_S5	83 RMT_GND_S5	67 CMN_GND	51 VDD_HS_DRV_2	35 VSW_S4	19 GND_S4	3 GND_S4
306 VSW_S9	290 VSW_S9	274 VDD_S9	258 PM_RESIN_N	242 VDD_S8	226 VSW_S8	210 GND_S8	194 GND_S7	178 VSW_S7	162 VDD_S7	146 VDD_S6	130 VSW_S6	114 GND_S6	98 GND_S5	82 VSW_S5	66 VDD_S5	50 CMN_GND	34 VDD_S4	18 VSW_S4	2 VSW_S4
305 VDD_S9	289 VDD_S9	273 VDD_S9	257 PM_PS_HOLD	241 VDD_S8	225 VSW_S8	209 GND_S8	193 GND_S7	177 VSW_S7	161 VDD_S7	145 VDD_S6	129 VSW_S6	113 GND_S6	97 GND_S5	81 VSW_S5	65 VDD_S5	49 CMN_GND	33 VDD_S4	17 VDD_S4	1 VDD_S4



PW6100-1 (A1)

PW6100-1 : Top View

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A	CHG_IN	CHG_IN	CHG_IN	CHG_MID	VSW_CHG	PGND_CHG	VPH_PWR	VBATT_PWR	OVP_SNS	OVP_DRV	VREG_BOB	VSW_BST_BOB	VSW_BCK_BOB	VDD_BOB	EUSB_EDM	USB_DM	USB_DP	VDD_XO_RF	VDD_XO_RF	VREG_XO
B	CHG_IN	CHG_IN	CHG_IN	CHG_MID	VSW_CHG	PGND_CHG	VPH_PWR	VBATT_PWR	SBU1	CHG_BYP_EN	VREG_BOB	PGND_BOB	VSW_BCK_BOB	VDD_BOB	EUSB_EDP	GND_EUSB	VDD_3V_EUSB	VREG_RF	VREG_XO	XTAL_OUT
C	CMN_GND	CMN_GND	CHG_MID	CHG_MID	VARB_CHG	BOOT_CAP	VPH_PWR	VBATT_SNS_M	SBU2	CC1_ID	VREG_BOB	VREG_BOB_SNS	VDD_PDPHY	VDD_BOB	VDD_1P2_EUSB	VDD_1P8_EUSB	VREG_1P6_EUSB	GND_XO	XTAL_OUT	XTAL_IN
D	CMN_GND	CMN_GND	VREG_CHG_BST	VREG_CHG_BST	REF_GND_CHG	CMN_GND	VPH_PWR	VBATT_SNS_P	CMN_GND	CC2	XOADC_GND	ADC_GND	AMUX_3	VREF_IX_2	VREF_IX_1	CLK5_LN	CLK4_LN	CLK1_RF	CMN_GND	KPD_PWR_N
E	VDD_S1	VDD_S1	RMT_GND_S1	CMN_GND	CMN_GND	CMN_GND	CMN_GND	CMN_GND	CMN_GND	XO_THERM	BATT_ID	RID	AMUX_2	AMUX_4	VREF_TXDAC	TEST_EN_VPP	CLK2_RF	CLK3_RF	VREG_CBST_SNS	VREG_CBST
F	VSW_S1	VSW_S1	VREG_S1	CMN_GND	VREG_L11	VREG_L17	VREG_1P2_SYS	VREG_1P8_SYS	CMN_GND	REF_BYP	BATT_THERM	AMUX_1	VREG_L19	VDD_L20	VDD_L22	VDD_L23	CMN_GND	CMN_GND	PGND_CBST	VSW_CBST
G	GND_S1	GND_S1	CMN_GND	VREG_L10	VDD_L10_L11	VDD_L17	VDD_L12_L13	VIN_1P8_SYS	XVDD_BYP	REF_GND	CMN_GND	VREG_L21	VDD_L19_L21	VREG_L20	VREG_L22	VREG_L23	SLEEP_CLK_IN	CMN_GND	CMN_GND	VDD_CBST
H	GND_S2	GND_S2	CMN_GND	VDD_L14	VREG_L14	VDD_L15_L16	VREG_L13	VREG_L12	CMN_GND	CMN_GND	PM_I2C_CLK	VDD_L30	VREG_L30	VREG_L7	VDD_L7_L26	VREG_L26	PM_SLEEP_CLK3	CMN_GND	GND_S11	GND_S11
J	VSW_S2	VSW_S2	VREG_S2	CMN_GND	VREG_L16	VREG_L24	VREG_L15	VREG_L28	PM_SPMI_DATA	PM_SPMI_CLK	PM_I2C_DATA	VPH_PWR_INFRA_1	VDD_L29	VREG_L29	VDD_L6	VREG_L6	PM_SLEEP_CLK2	VREG_S11	VSW_S11	VSW_S11
K	VDD_S2	VDD_S2	RMT_GND_S2	PM_GPIO_17	VDD_L25	VDD_L24	VDD_L28	VDD_L18	PM_GPIO_18	PM_GPIO_16	CMN_GND	CMN_GND	VREG_L9	VDD_L9	VDD_L8_L27	VREG_L27	PM_SLEEP_CLK1	RMT_GND_S11	VDD_S11	VDD_S11
L	VDD_S3	VDD_S3	RMT_GND_S3	PM_GPIO_04	PM_GPIO_01	VREG_L25	CMN_GND	VREG_L18	PM_GPIO_15	PM_GPIO_14	CMN_GND	CMN_GND	VREG_L4	VDD_L4_L5	VDD_L1	VREG_L8	PMIC_RB	RMT_GND_S10	VDD_S10	VDD_S10
M	VSW_S3	VSW_S3	VREG_S3	PM_GPIO_03	PM_GPIO_02	CMN_GND	PM_GPIO_08	CMN_GND	PM_GPIO_13	PM_GPIO_12	CMN_GND	CMN_GND	VREG_L5	VDD_L2	VDD_L3	VREG_L1	PM_PON_RESET_N	VREG_S10	VSW_S10	VSW_S10
N	GND_S3	GND_S3	VREG_S4	RMT_GND_S4	PM_GPIO_05	PM_GPIO_06	PM_GPIO_07	PM_GPIO_09	PM_GPIO_10	PM_GPIO_11	CMN_GND	CMN_GND	VREG_L2_SNS	VREG_L2	VREG_L3	VREG_L3_SNS	RMT_GND_S9	VREG_S9	GND_S10	GND_S10
P	GND_S4	GND_S4	VSW_S4	VDD_HS_DRV_2	CMN_GND	RMT_GND_S5	VREG_S5	VPH_PWR_INFRA_2	VREG_S6	RMT_GND_S6	RMT_GND_S7	VREG_S7	CMN_GND	VREG_S8	RMT_GND_S8	FAULT_N	VDD_HS_DRV_1	VSW_S9	GND_S9	GND_S9
R	VSW_S4	VSW_S4	VDD_S4	CMN_GND	VDD_S5	VSW_S5	GND_S5	GND_S6	VSW_S6	VDD_S6	VDD_S7	VSW_S7	GND_S7	GND_S8	VSW_S8	VDD_S8	PM_RESIN_N	VDD_S9	VSW_S9	VSW_S9
T	VDD_S4	VDD_S4	VDD_S4	CMN_GND	VDD_S5	VSW_S5	GND_S5	GND_S6	VSW_S6	VDD_S6	VDD_S7	VSW_S7	GND_S7	GND_S8	VSW_S8	VDD_S8	PM_PS_HOLD	VDD_S9	VDD_S9	VDD_S9



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Page 1

WRD6100 SKU1 vs SKU2 Comparison

Feature Category	SKU1 Details	SKU2 Details
SoC	SW6100-0/1	SW6100-0/1/**2**
PMIC	**PMW6100**	**PM6100-0/1**
Display	1.39”, Circle, AMOLED LTPS 454 x 454, 5 ~ 45 Hz	1.39”, Circle, AMOLED LTPS 454 x 454, 5 ~ 45 Hz
Cameras	Front facing camera, 12MP FF, CPHY, sensor: IMX681	**Not supported**
Memory	**64 GB eMMC5.1 + 4 GB LP5X** 512 MB 266 MHz/16bit PSRAM	**32 GB eMMC5.1 + 2 GB LP5X (Only with SW6100-2)** 512 MB 266 MHz/16bit PSRAM
Battery & Charging	470 mAh/2.5 V ~ 4.48 V, 2C charging/discharging, Wireless charging (CPS4019A), USB2.0 (through debug FPC)	470 mAh/2.5 V ~ 4.48 V, 2C charging/discharging, Wireless charging (CPS4019A), USB2.0 (through debug FPC)
Sensors	Accelerometer, Pressure, Ambient Light Sensor, E-compass, Body Skin Temp, HRM (PPG+ECG), **Electroneurography (ENG)–Not PoR**	Accelerometer, Pressure, Ambient Light Sensor, E-compass, Body Skin Temp, HRM (PPG+ECG)
WWAN Configurations	Global bands, 5G Sub6, **ET, DRX**, UHB	Global bands, 5G Sub6, **APT**, UHB
Audio	3x DMICs, 1x smart speaker, 1x LRA haptic motor	3x DMICs, 1x smart speaker, 1x LRA haptic motor
Connectivity	Wi-Fi 6 (802.11ax) 1x1, UWB channels 5, 9, Bluetooth 5.4+LE, NFC+ eSE + eSIM: ST54L, GNSS Dual band (L1, L5), Satellite NB-NTN (B255, B256, B23)	Wi-Fi 6 (802.11ax) 1x1, UWB channels 5, 9, Bluetooth 5.4+LE, NFC+ eSE + eSIM: ST54L, GNSS Dual band (L1, L5), Satellite NB-NTN (B255, B256, B23)

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NOTE: Electroneurography (ENG) sensor is Not PoR. For more details contact QTI.

Section 3

WRD6100 Reference Platforms & Reference Designs

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WRD6100 Reference Documents

SW6100 Variant	DCN	Description
SW6100-1 SKU1	DP25-83951-200	DESIGN PACKAGE, SW6100-1 , WRD6100 REFERENCE DESIGN FILES
SW6100-2 SKU2	DP25-83952-200	DESIGN PACKAGE, SW6100-2 , WRD6100 REFERENCE DESIGN FILES
SW6100-1 SKU1	DP25-83951-230	DESIGN PACKAGE, WRD6100 SKU1, SMART WATCH, MECHANICAL HARDWARE, MECH PLASTIC & STAMPED METAL CAD
SW6100-2 SKU2	DP25-83952-230	DESIGN PACKAGE, WRD6100 SKU2, SMART WATCH, MECHANICAL HARDWARE, MECH PLASTIC & STAMPED METAL CAD

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WRD6100 Reference Platforms

SKU	MCN	Revision	SoC Device	PMIC Device	Audio PA	RF/RFFE	ePOPMemory	PSRAM	Display	Camera	WCN
SKU1	65-82482-1013	EVT1.1	SW6100 8x10 MEP	PWM6100 Short V1.0	PWM6100 Short V1.0	SDR125 + QPA6590 + QFM6515 + QPM7477 + QDM5302 + QET5100	Kinston 64 GB eMMC + 4 GB LPDDR5x	64 MB, 200 MHz	1.39” AMOLED	Front 12 MP	QCC6100 V1.0
	65-82482-1014	EVT1.1	SW6100 8x10 RDL	PWM6100 Short V1.0	PWM6100 Short V1.0	SDR125 + QPA6590 + QFM6515 + QPM7477 + QDM5302 + QET5100	Kinston 64 GB eMMC + 4 GB LPDDR5x	64 MB, 200 MHz	1.39” AMOLED	Front 12 MP	QCC6100 V1.0
	65-82482-2005	DVT1.0	SW6100 8x10 MEP	PWM6100 Short V2.0	PWM6100 Short V2.0	SDR125 + QPA6590 + QFM6515 + QPM7477 + QDM5302 + QET5100	Kinston 64 GB eMMC + 4 GB LPDDR5x	64 MB, 200 MHz	1.39” AMOLED	Front 12 MP	QCC6100 V2.0
	65-82482-2007	DVT1.0	SW6100 8x10 RDL	PWM6100 Short V2.0	PWM6100 Short V2.0	SDR125 + QPA6590 + QFM6515 + QPM7477 + QDM5302 + QET5100	Kinston 64 GB eMMC + 4 GB LPDDR5x	64 MB, 200 MHz	1.39” AMOLED	Front 12 MP	QCC6100 V2.0
	65-82482-2016	DVT1.1	SW6100 8x10 MEP	PWM6100 Short V2.0	PWM6100 Short V2.0	SDR125 + QPA6590 + QFM6515 + QPM7477 + QDM5302 + QET5100	Kinston 64 GB eMMC + 4 GB LPDDR5x	64 MB, 266 MHz	1.39” AMOLED	Front 12 MP	QCC6100 V2.0
	65-82482-2018	DVT1.1	SW6100 8x10 MEP	PWM6100 Short V2.0	PWM6100 Short V2.0	SDR125 + QPA6590 + QFM6515 + QPM7477 + QDM5302 + QET5100	Kinston 64 GB eMMC + 4 GB LPDDR5x	65 MB, 266 MHz	1.39” AMOLED	Front 12 MP	QCC6100 V2.0
	65-82482-2017	DVT1.1	SW6100 8x10 RDL	PWM6100 Short V2.0	PWM6100 Short V2.0	SDR125 + QPA6590 + QFM6515 + QPM7477 + QDM5302 + QET5100	Kinston 64 GB eMMC + 4 GB LPDDR5x	66 MB, 266 MHz	1.39” AMOLED	Front 12 MP	QCC6100 V2.0

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2026-02-05 9:18 GMT-08:00 Kaushik Vaidyanathan (kaushikv@qti.qualcomm.com)

WRD6100 Reference Platforms (cont.)

SKU	MCN	Revision	SoC Device	PMIC Device	Audio PA	RF/RFFE	ePOPMemory	PSRAM	Display	Camera	WCN
SKU2	65-82483-1011	EVT1.1	SW6100 8x13 MEP	PW6100 FC V1.0	WSA8855 2.0	SDR125 + QPA6590 + QFM6515 + QET6105	SS 32GB eMMC + 2GB LPDDR5x	63 MB, 200 MHz	1.39” AMOLED	-	QCC6100 V1.0
	65-82483-2003	DVT1.0	SW6100 8x13 MEP	PW6100 FC V2.0	WSA8855 2.0	SDR125 + QPA6590 + QFM6515 + QET6105	SS 32GB eMMC + 2GB LPDDR5x	64 MB, 200 MHz	1.39” AMOLED	-	QCC6100 V2.0
	65-82483-2012	DVT1.1	SW6100 8x13 MEP	PW6100 FC V2.0	WSA8855 2.0	SDR125 + QPA6590 + QFM6515 + QET6105	SS 32GB eMMC + 2GB LPDDR5x	64 MB, 266 MHz	1.39” AMOLED	-	QCC6100 V2.0

Note: Some of the configurations mentioned in the above table may not be readily available for ordering, please contact Qualcomm Sales team for more information.

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Section 4

WRD6100 SKU1

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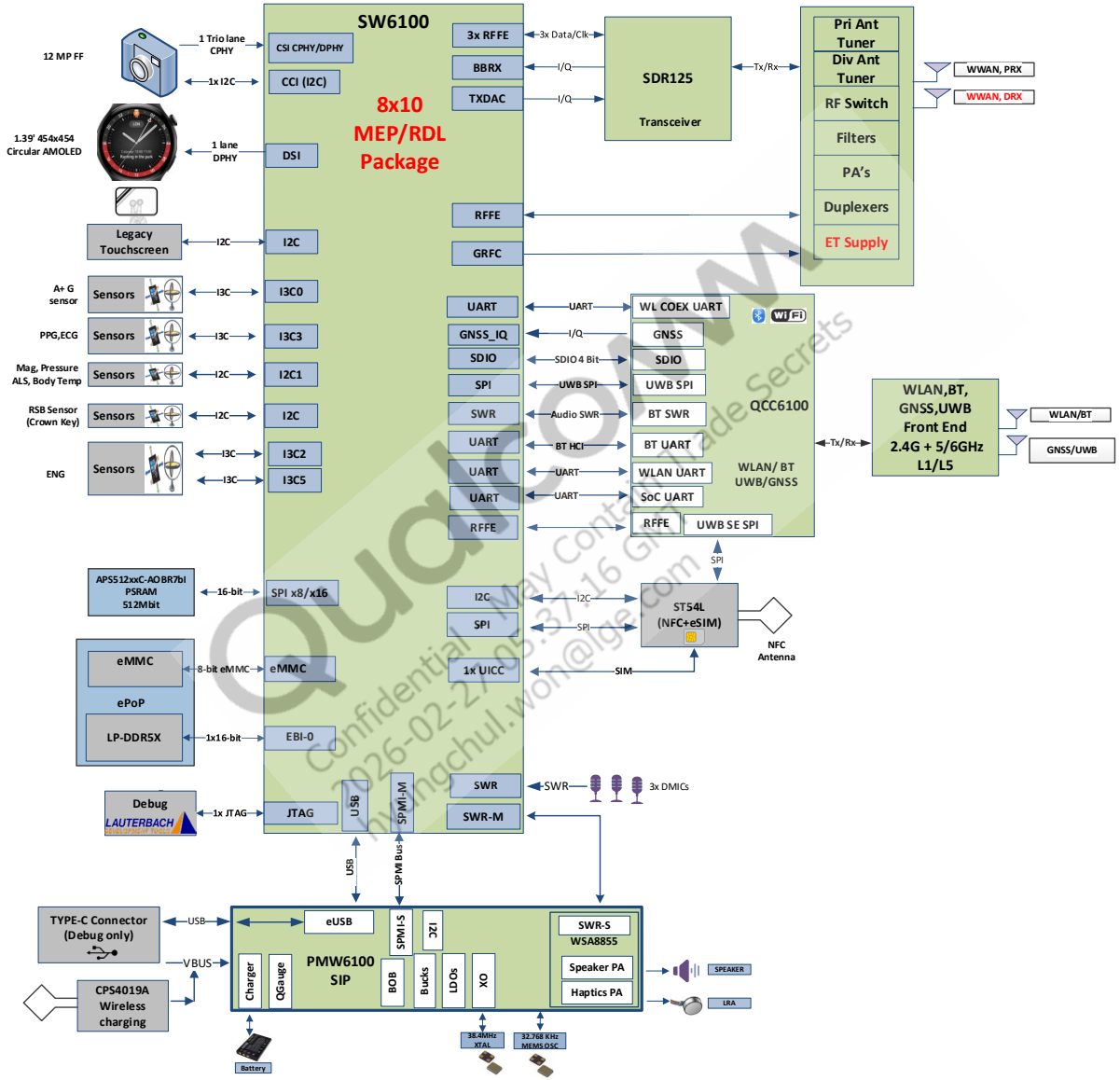
WRD6100 SKU1 Block Diagram



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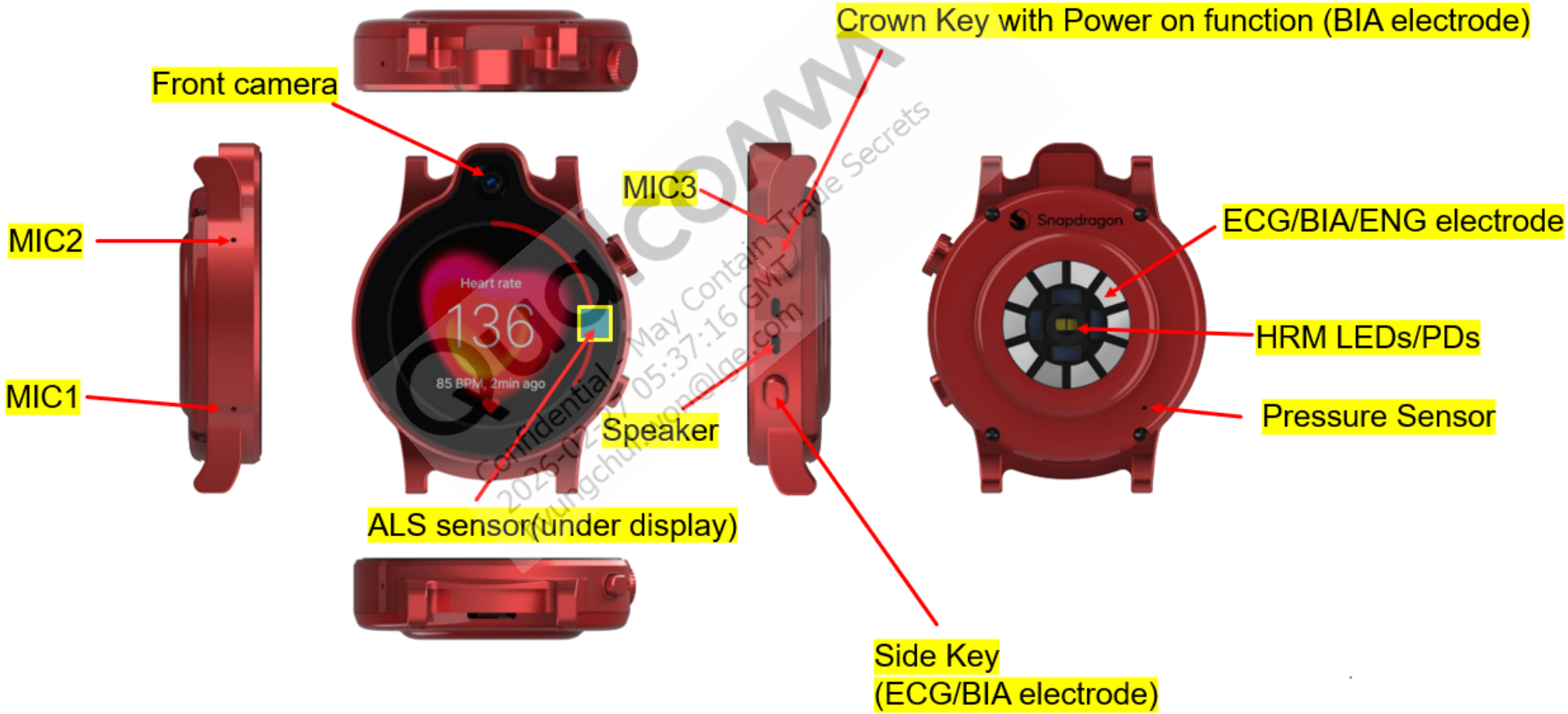
WRD6100 SKU1 ID

- Color: Snapdragon® Red
- Size: 47.6 x 47.6 x 14.0 mm
- 54.8 mm with camera bump



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WRD6100 SKU1 Features

- Display:
  - 1.39”, Circle, AMOLED LTPS 454 x 454, 5 ~ 45 Hz
- Cameras:
  - Front facing camera, 12MP FF, CPHY, sensor: IMX681
- Memory:
  - 64 GB eMMC5.1 + 4 GB LP5X
  - 512 MB 266 MHz/16 bit PSRAM
- Battery & charging:
  - 470 mAh/2.5 V ~ 4.48 V, 2C charging/discharging
  - Wireless charging (CPS4019A)
  - USB2.0 (through debug FPC)
- Sensors:
  - Accelerometer
  - Pressure
  - Ambient Light Sensor
  - E-compass
  - Body Skin Temp
  - HRM (PPG+ECG)
  - Electroneurography (ENG) (Optional)
- WWAN configurations:
  - Global bands, 5G Sub6, ET, DRX, UHB
- Audio:
  - 3x DMICs
  - 1x smart speaker
  - 1x LRA haptic motor
- Connectivity:
  - Wi-Fi 6 (802.11ax) 1x1
  - UWB channels 5, 9
  - Bluetooth 5.4+LE
  - NFC+ eSE + eSIM: ST54L
  - GNSS Dual band (L1, L5)
  - Satellite NB-NTN (B255, B256, B23)
  - **Physical SIM Card not supported**

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Supported Sensors

Sensors	Part number
Accelerometer/gyro	LSM6DSV32X
Pressure	BMP585
HRM/ECG	MAX86178ENJ+
ALS sensor	OPT3004DNPR
RSB sensor	PAT9401E1
Magnetometer	AK09920C
Body temperature sensor	CT7117(x2)
ENG sensor	STENG1AX(x4) (Not PoR)

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WRD6100 Platforms Key chips – SKU1

	SoC	WiFi-BT	RF							
Processor, RFFE, Wi-Fi	SW6100 MEP/RDL SKU1	QCC6100	SDR125	QPA6590	QFM6515	QPM7477	QDM5302	QET5100M	QAT5625 X 2	QAT5655

	Main PMIC	Display PMIC	Wireless charge
PMIC	PMW6100	AAT1556	CPS4019 A
Memory/NFC	ePOP eMMC+LP5X	PSRAM APS256XXN	NFC ST54L (with eSIM)
Audio	PMW6100 Audio PA	DMICs QMP1000 X 3	

	1.39' Circle	Front CAM
DISPLAY & Camera	Display DDIC RM6D030	12MP IMX681

	ACCE+GYRO	MAGNETO	HRM/ECG	RSB	ALS	PRESSURE	Body temp	ENG
Sensors	LSM6DSV32 X ST	AK09920C AKM	MAX86178E NJ ADI	PAT9401E1 PIXART	OPT3004 TI	BMP585 BOSCH	CT7117 X 2 Sensyslink	STENG1AX X 4 ST

QCOM Chip

3rd Chip

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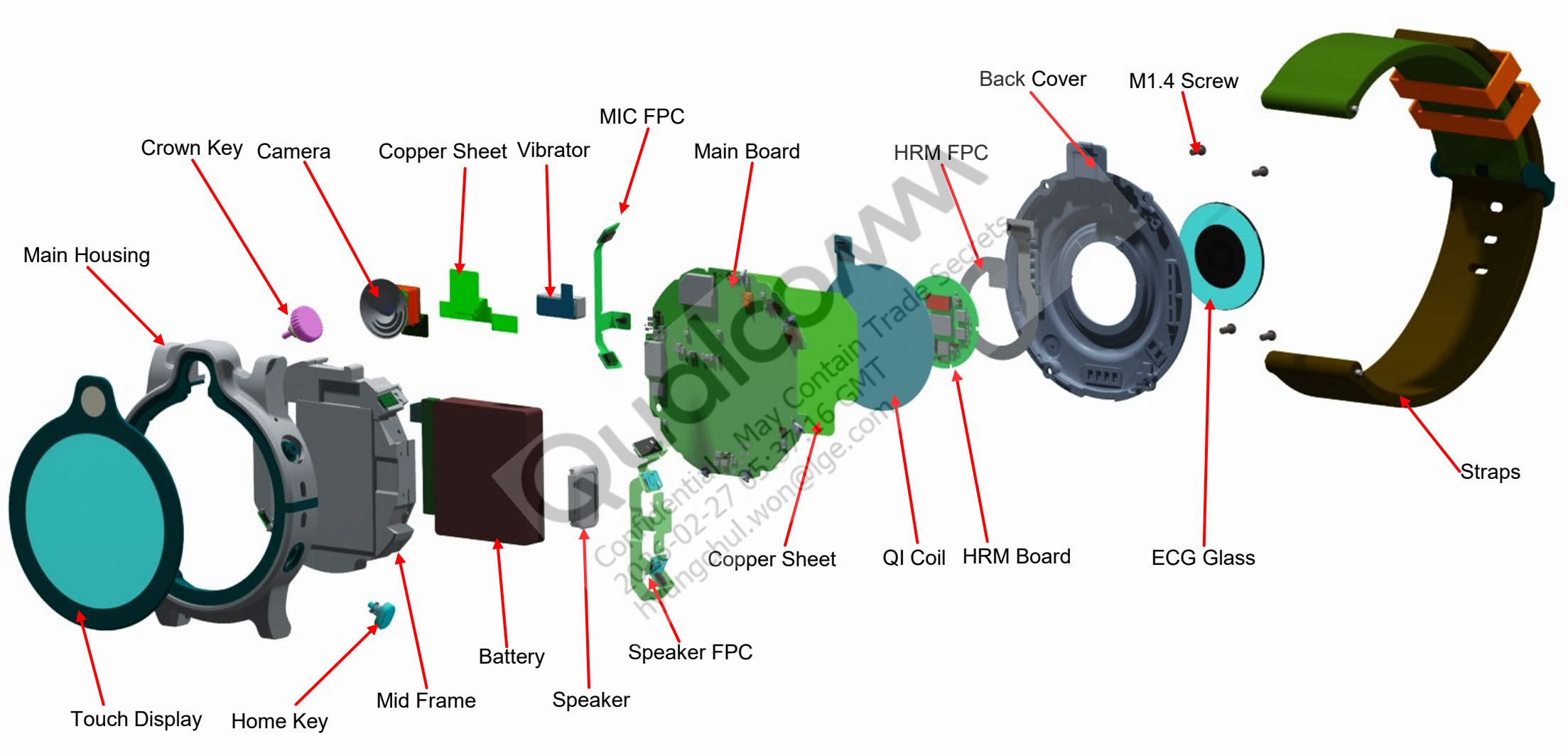
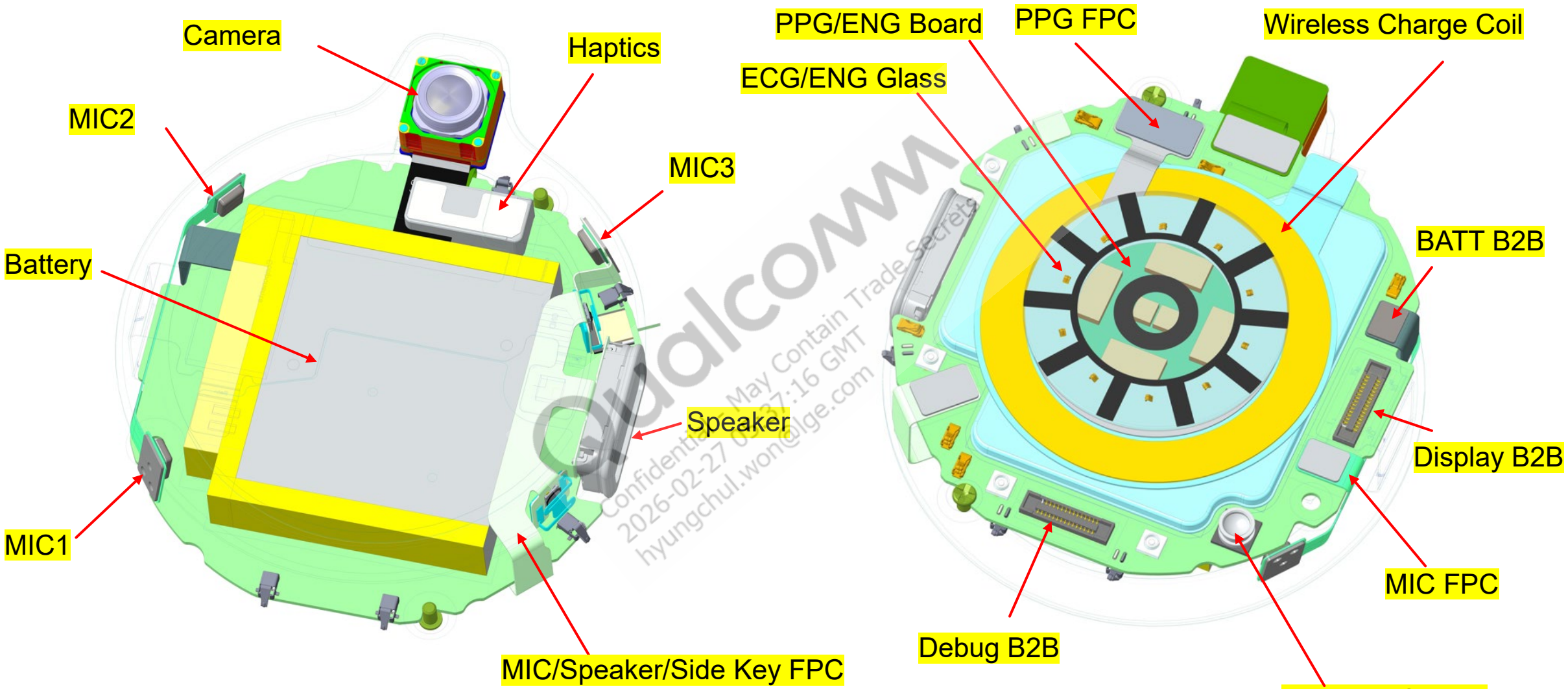
WRD6100 SKU1 Exploded View



WRD6100 SKU1 Mechanical Design – Placement



Top (Display Side)

Bottom

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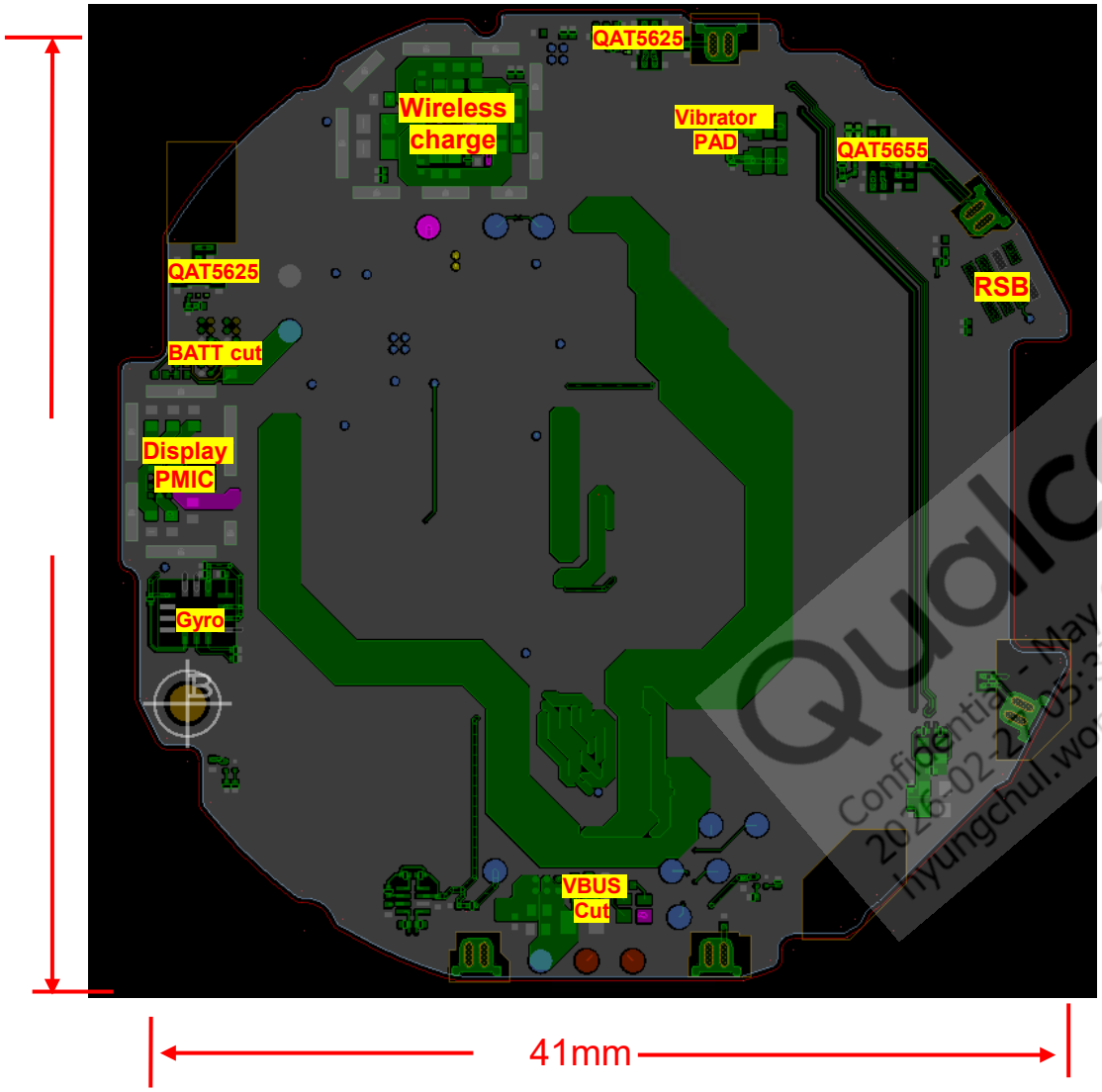
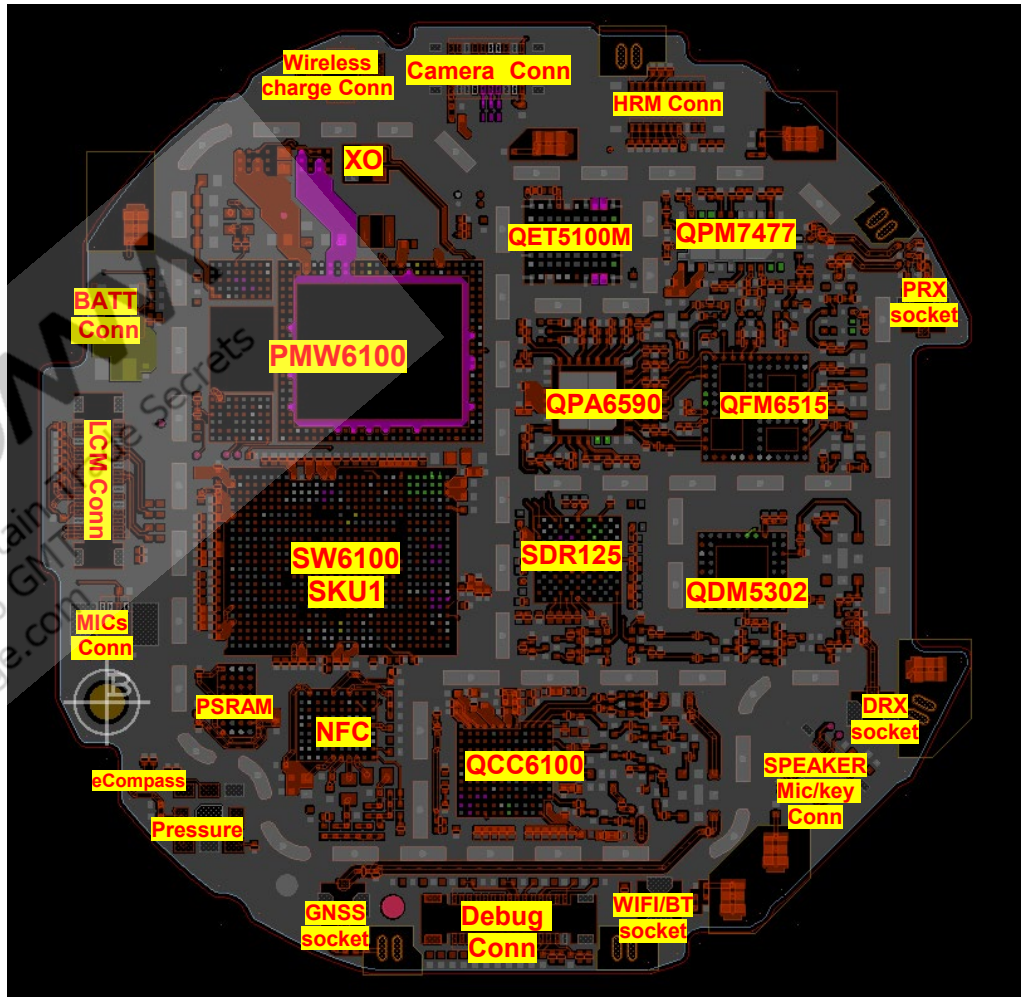
Main Board Overall Dimensions and Placement for SKU1



Top Side (Face To Display)



Bottom Side

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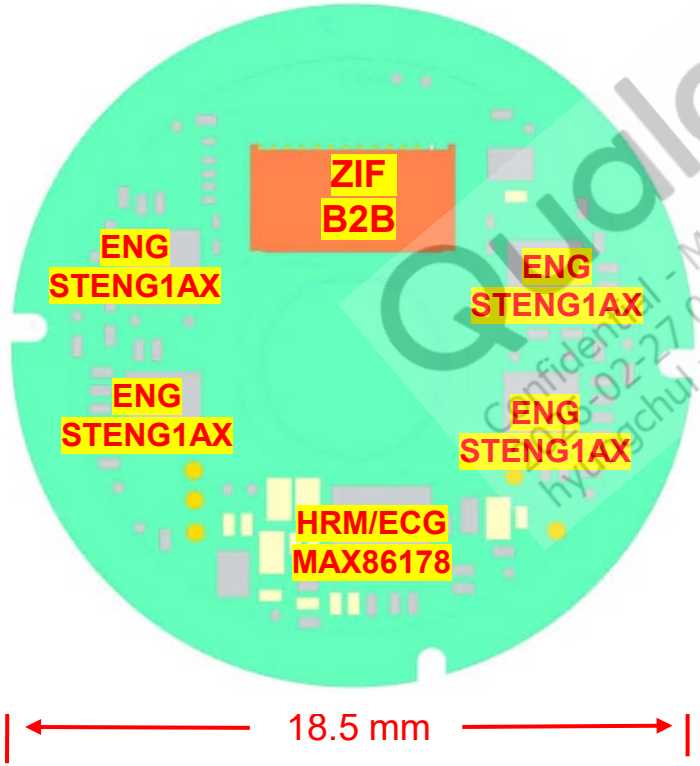
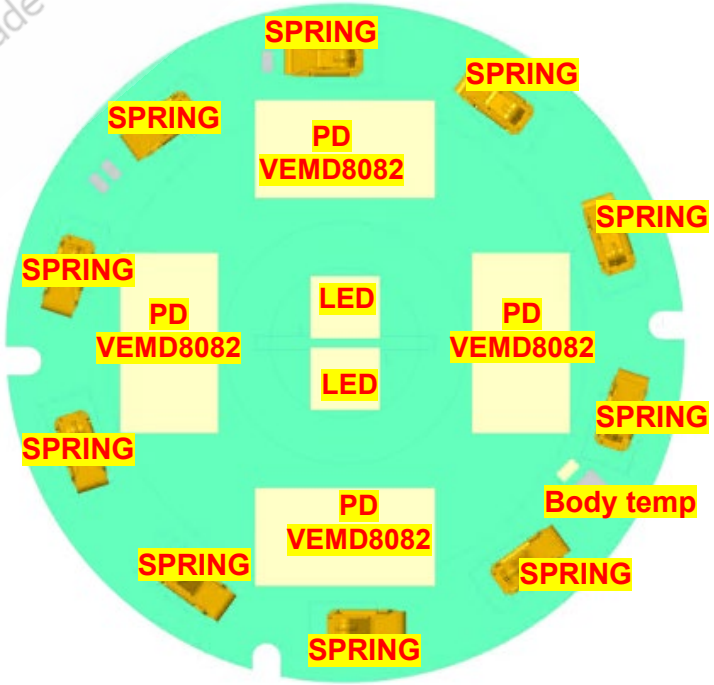
HRM Board Overall Dimensions and Placement SKU1

- 20-79382-H1 board
  - PPG/ECG IC: MAX86178ENJ+
  - 2 x LEDs for transmitting (Green_Red_IR)
  - 4 x PDs for receiving
  - 1 x body temperature sensor
- 20-79382-H2 board
  - 4 x STENG1AX ENG sensor (Not PoR)

Top Side (Face To Display)



Bottom Side



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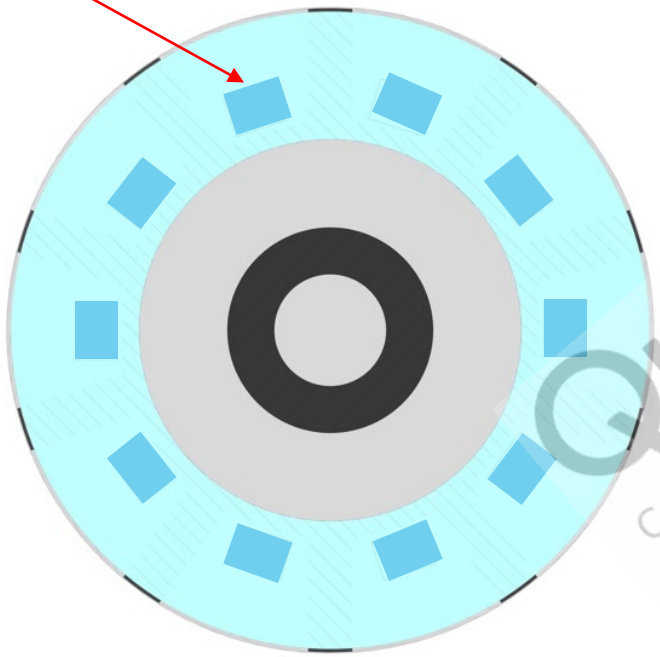
Mechanical PPG/ECG/ENG Glass SKU1

- PPG/ECG/ENG glass

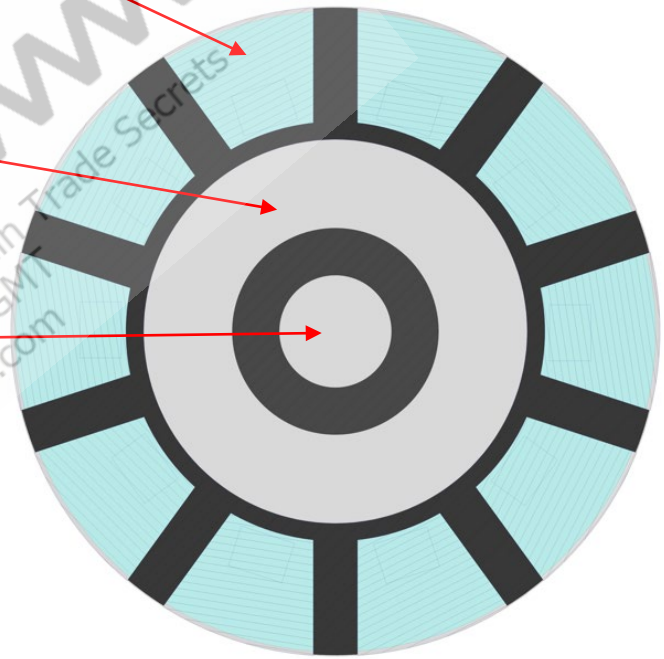
Electrode contacts



10 x electrodes for ENG config,2 x electrodes for ECG config

Photo diode window

LED window



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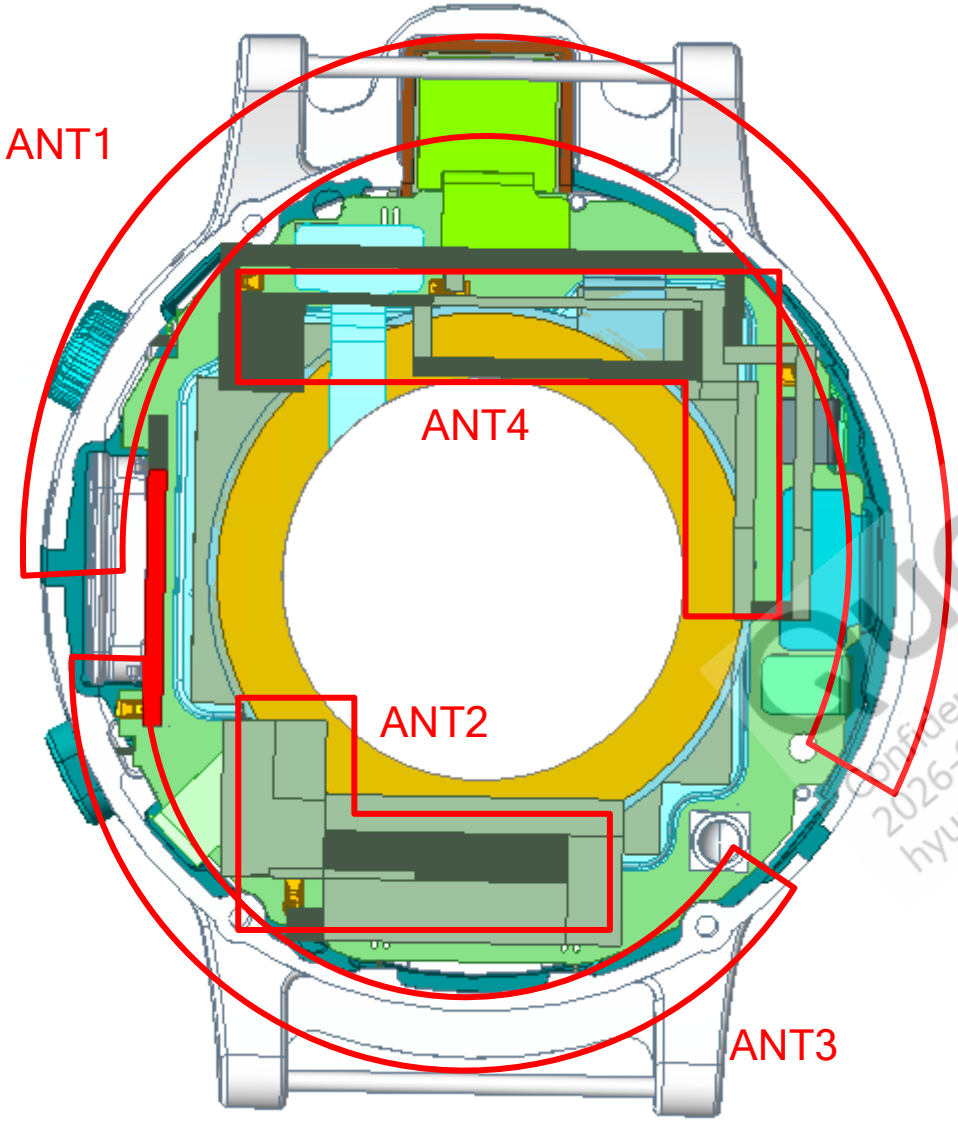
Overall Antenna System SKU1



ANT	Function	Frequency	Antenna Type	Tuner
ANT1	L/M/HB/ Tx0 NTN	703~960 MHz 1525~2700 MHz	Metal ring	QAT5655 QAT5625
ANT2	WIFI(2.4/5/6G) BT	2400~2500 MHz 5100~7125 MHz	FPC	/
ANT3	GNSS L1/L5 UWB	1174 MHz 1559~1606 MHz 6240~8240 MHz	Metal ring	/
ANT4	M/HB Rx0 UHB Tx0	1805~2700 MHz 3300~4200 MHz	FPC	QAT5625

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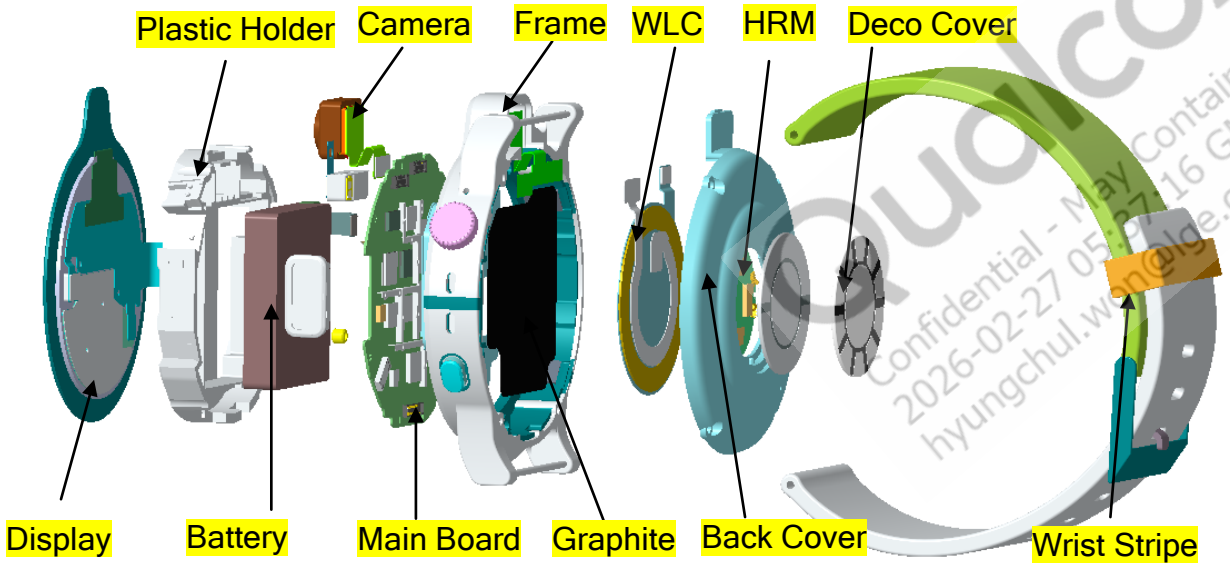
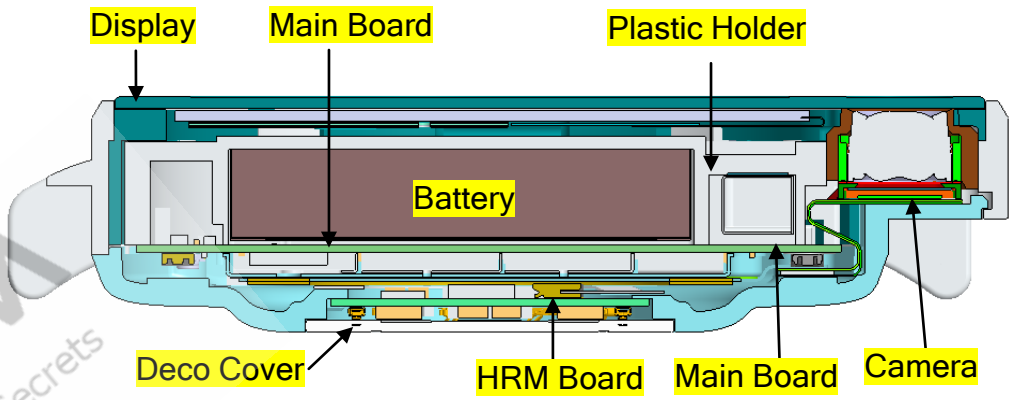
Thermal Solution and Stack-up SKU1

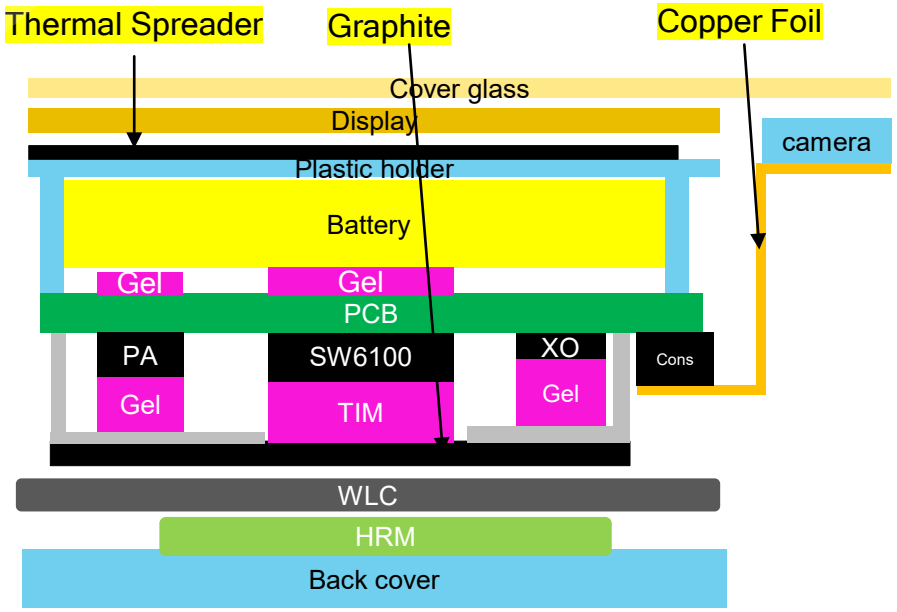
- There is no graphite inside back cover due to antenna clearance.
- Keep gap between main board and battery to prevent battery expansion risk, just add gel opposite area of SW6100 and QPA6590.



Explode View



Section View



Thermal Stack-up

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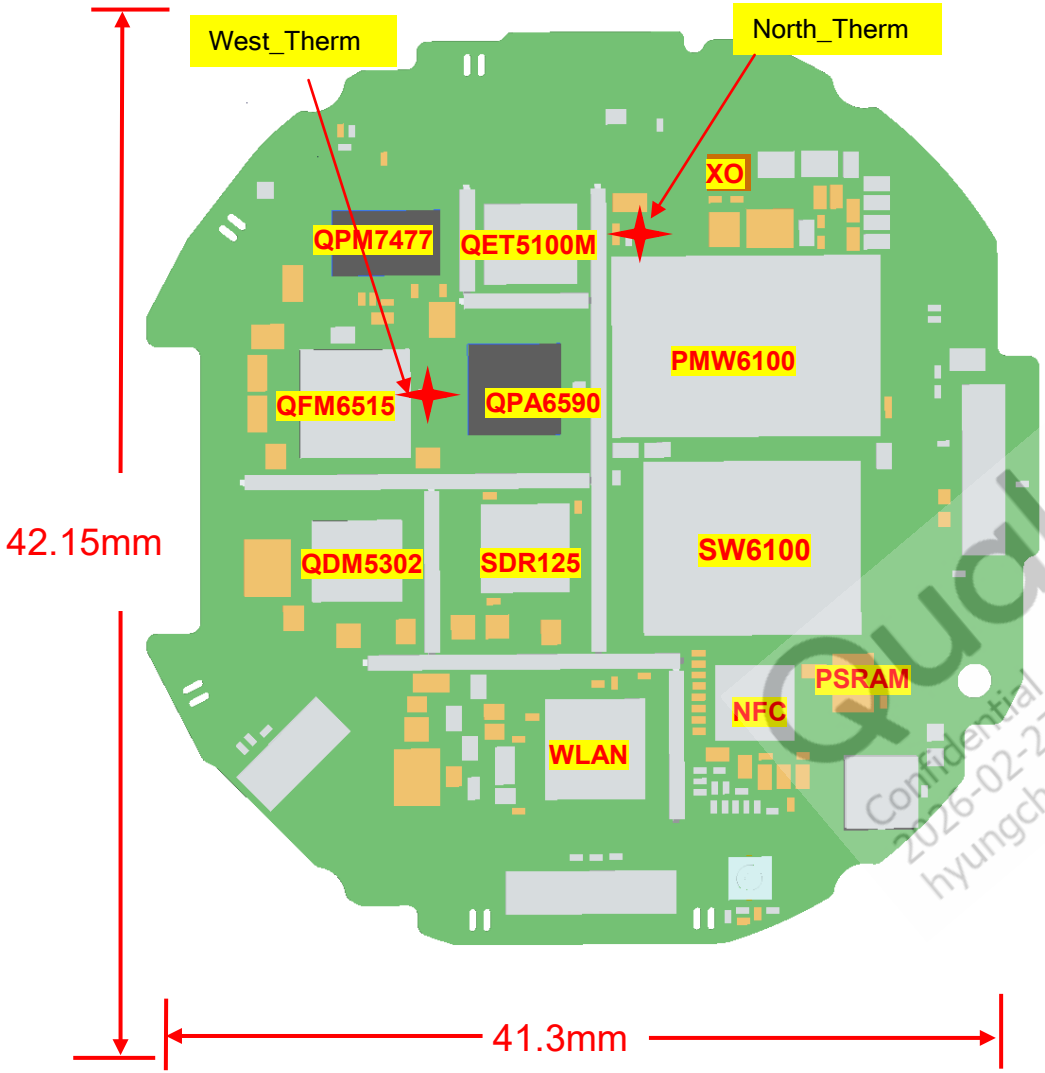
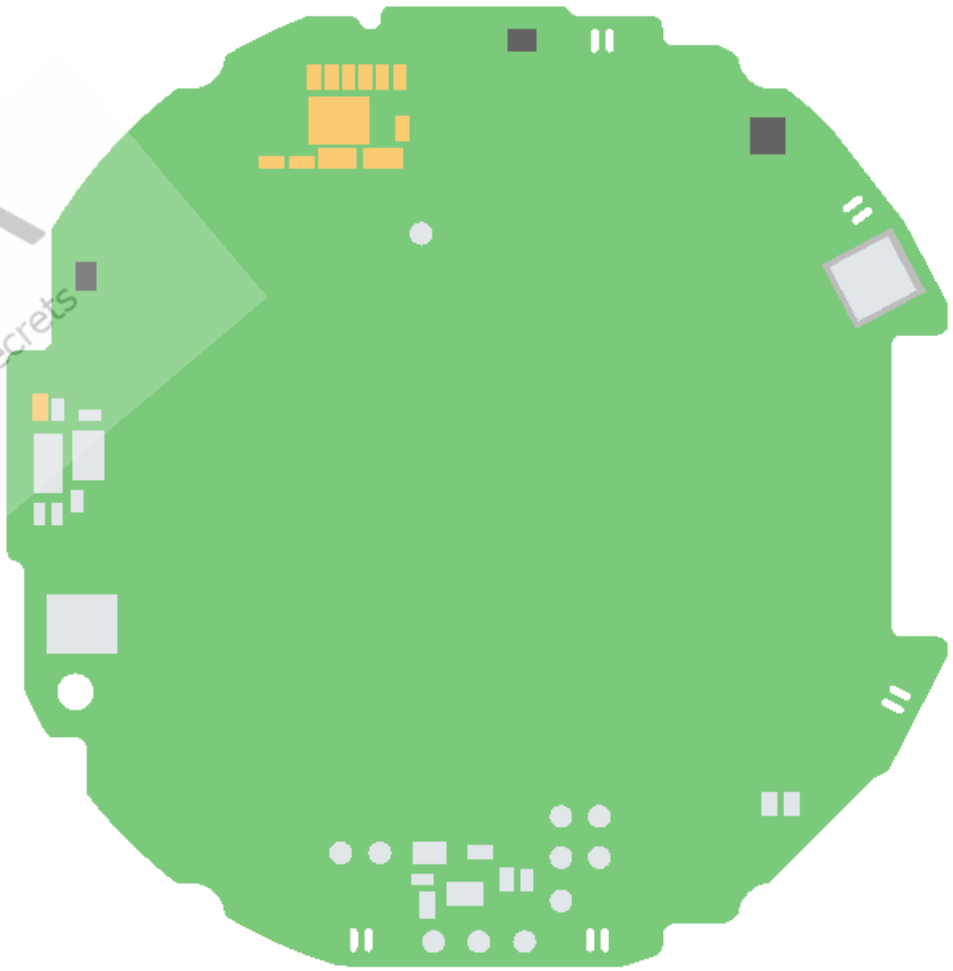
Thermistor PlacementSKU1



Bottom (Back Cover)



Top (Display)

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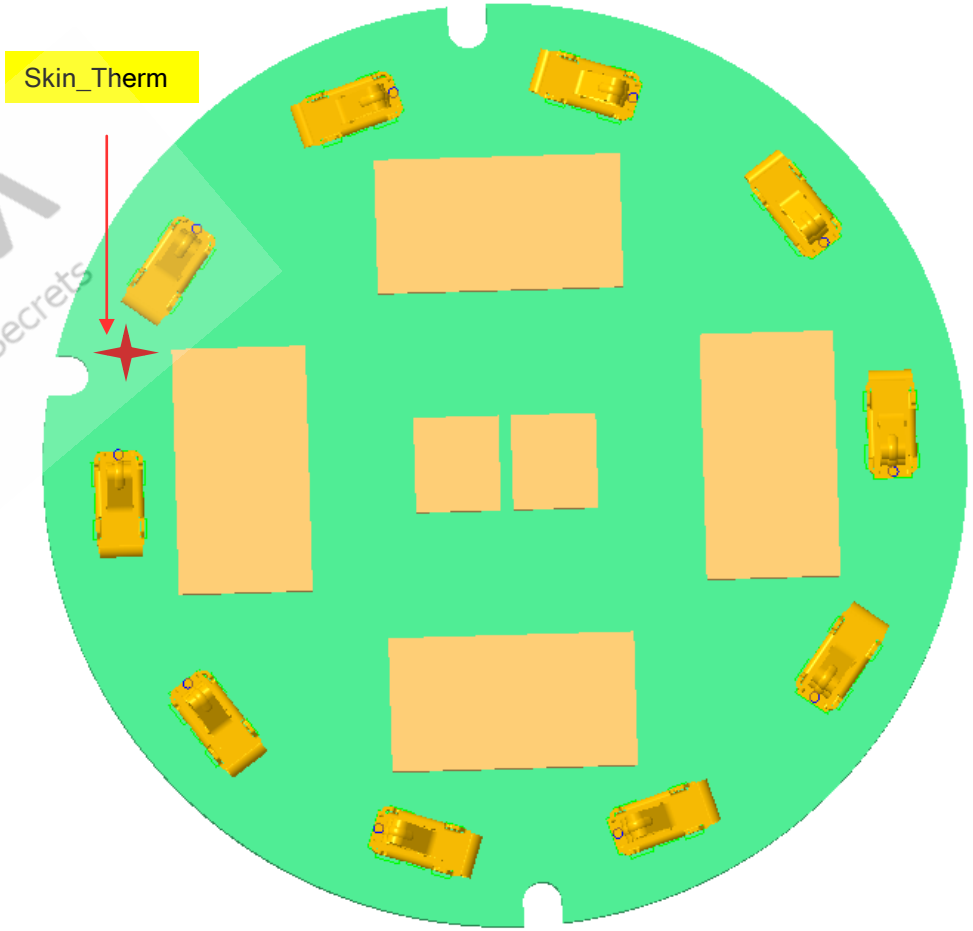
Thermistor Placement SKU1

- There are 3 external thermistors in design.

PMIC and Pin	Thermistor Function
AMUX_1	NORTH_THERM (GNSS)
AMUX_2	WEST_THERM (RF PA)
AMUX_3	SKIN_THERM (It is on HRM board, close to SKIN)
AMUX_4	Boot option



HRM SUB PCB

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WRD6100 SKU1 Key Component List

Category	Function	Description	Qty	WRD6100 SKU1 MEP/RDL 8x10 + PMIC SIP Config1	Vendor
Chipset	MSM™ Chipsets	Main chip	1	SW6100 MEP/RDL 8x10	Qualcomm
	PMIC	Core PMIC	1	PMW6100	Qualcomm
Display	AMOLED display	1.39" circle LTPS	1	Module: ATV-D0139MAMBCA6_V0 AUO 1.39" 454*454 DDIC RM6D030 (Raydium)	ANTAVIC
	Touch	InCell touch	1	Module: ATV-D0139MAMBCA6_V0 DDIC RM6D030 (Raydium)	ANTAVIC
Camera	Front camera	Front module 12 MP FF	1	Module:CM24BFA02 12MP FF,CPHY Sensor:IMX681	LARGAN
Sensors	G-sensor+Gyro sensor	6-axis	1	LSM6DSV32X	ST
	Magnetometer	3-axis	1	AK09920C	AKM
	ALS	Ambient light sensor	1	OPT3004DNPR	TI
	Pressure	Pressure sensor	1	BMP585	Bosch
	Body skin temperature	Body skin temperature	1	CT7117AJ4R	Sensyslink
			1	CT7117EJ4R	Sensyslink
	HRM/ECG sensor	Heart rate sensor Electrocardiogram sensor	1	MAX86178ENJ+	ADI
	RSB sensor	Crown key rotation sensor	1	PAT9401E1	PixArt
	Photodiode	Photodiode	4	VEMD8082	Vishay
	Multi Chip LED	Multi chip LED	2	SFH7016	OSRAM
	ENG	ENG (electroneurography)	4	STENG1AX	ST
Memory	ePOP memory	eMMC + LPDDR5X	1	64EP32-M5BTB9M-GA03 ePOP (64 GB eMMC5.1 + 4 GB LP5x, 4.2 GHz) 8 mm x 9.5 mm, 0.7 mm max height	Kingston
	PSRAM	SPI PSRAM	1	APS512XXC-AOB7NI-WB x16 512Mbit 1.2V 266 MHz DDR	AP Memory
Wireless charging	Charging	Wireless charging IC	1	CPS4019A	Convenient Power
NFC	NFC	NFC	1	ST54L03AEL9CCJ7 (Integrate eSIM)	ST MICROELECTRONICS
AMOLED Power IC		AMOLED power IC	1	AAT1556	ADVANCED ANALOG
Clock	Crystal	38.400 MHz	1	EXS00A-CS12755 1.6mmX1.2mm	NDK
	XO	32.768 KHz	1	SIT1566A-JE-0001	Si Time

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WRD6100 SKU1 Key Component List (cont.)

Category	Function	Description	Qty	WRD6100 SKU1 MEP/RDL 8x10 + PMIC SIP Config1	Vendor
Audio	MIC	Digital MEMS MIC	3	QMP1000	Qualcomm
	Speaker	Speaker	1	FG-1206H2S-6-P	Forgrand
Vibrator	LRA	AC linear resonant actuator	1	BLM0410G-FP249-0RUAJ	Baolong
Battery	Battery	Lithium-ion battery	1	AEC492627 Si-Anode/450mAH/2C charger/discharge 2.5V Cut-off	AEC
Cellular	Chipset	RFIC	1	SDR125	
	QAT	Aperture Tuner	2	QAT5625	Qualcomm
		Impedance Tuner	1	QAT5655	
	PA/ASM	4G LMH-PA	1	QPA6590	
		LMH L-FEMID	1	QFM6515	
		n77/78/79-LPAF	1	QPM7477	
	QDM	LMH L-DivFEM	1	QDM5302	
	QET	QET	1	QET5100	
	LNA	eLNA	2	MXD8921HGW	Maxscend
	Switch	DPDT	1	CXA4482GC-E-T9	Sony
		SP2T	2	BGS12WN6E6327XTSA1	Infineon
		SP2T	4	QM12312	Qorvo
	N-Plexer	Triplexer	1	TPX255950MT-7089A1	Murata
		Diplexer	1	LFD18829MMP1E441	Murata
	Filter	N255/266 TX SAW	1	B7585	RF360
		N255/266 RX SAW	2	B7586	RF360
		LB LPF	1	DEA160960LT-5059A1	TDK
		5G WL Jammer FL	1	DEA162690LT-5064A1	TDK
		BT Notch	2	BJ58A (ES)	RF360
GNSS	Triplexer	GNSS+B255+UWB	1	TPX259236MT-7343A1(ES)	TDK
	GNSS BPF	L1 BPF	2	B39162B7504L210	RF360
		L5 BPF	2	B39122B7505L210	RF360
	GNSS eLNA	L1 eLNA	1	BGA123N6	Infineon
		L5 eLNA	1	BGA125N6	Infineon
WLAN	Chipset	WCN	1	QCC6100	Qualcomm
	Diplexer	Diplexer	1	DPX107250DT-6130B1	TDK
	5G filter	BPF	1	DEA146138BT-2413B1	TDK
	2.4G filter	BPF	1	B39242B7530L210	RF360
	UWB filter	BPF	1	DEA167239BT-2404B1(ES)	TDK
	1P2T switch	Switch	1	BGS12WN6E6327XTSA1	Infineon

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Section 5

WRD6100 SKU2

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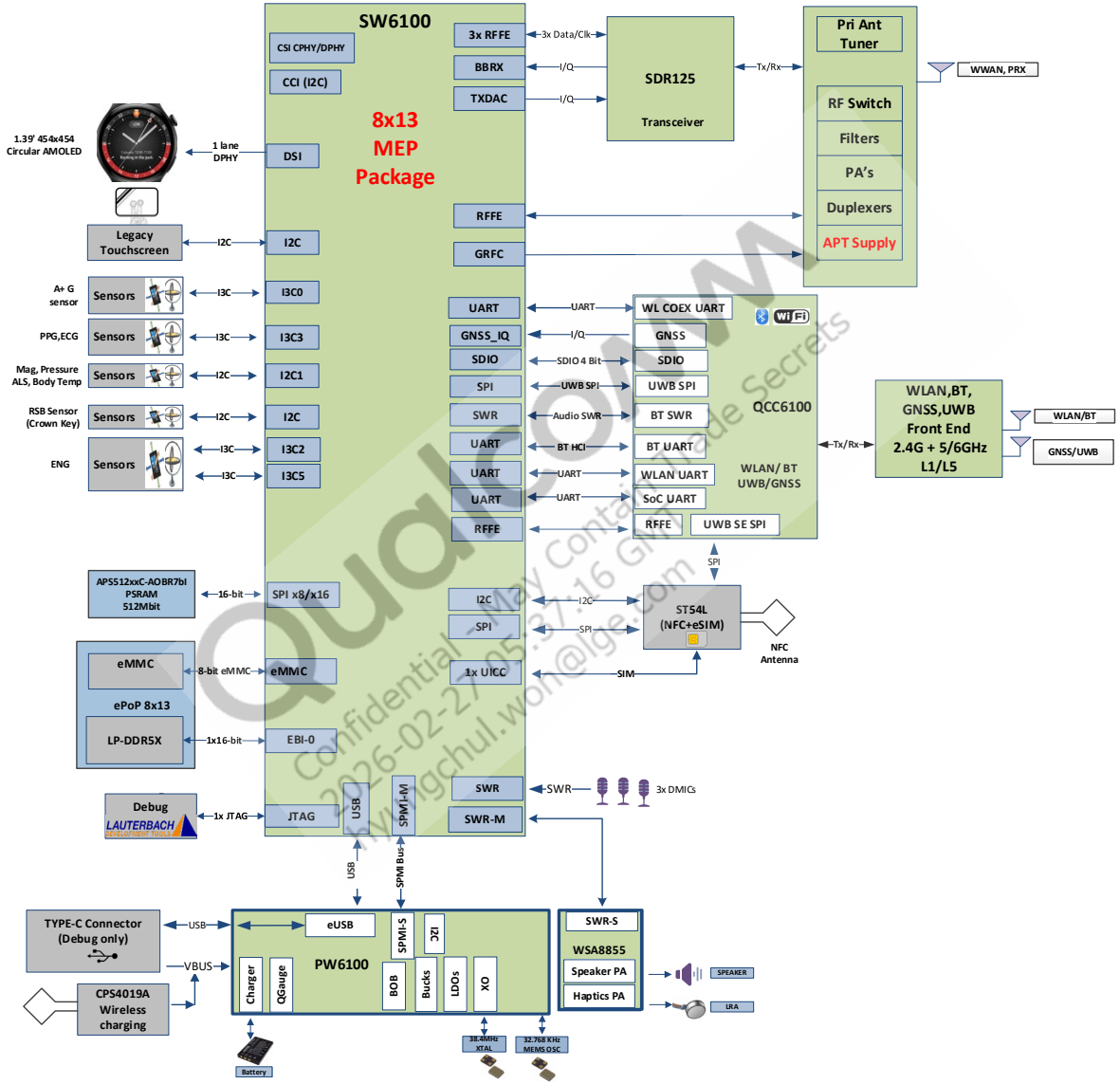
WRD6100 SKU2 Block Diagram



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WRD6100 SKU2 ID

- Color: Snapdragon Red
- Size: 47.6 x 47.6 x 14.0 mm
- 54.8 mm with camera bump

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WRD6100 SKU2 Features

- Display:
  - 1.39”, Circle, AMOLED LTPS 454 x 454, 5 ~ 45 Hz
- Memory
  - 32 GB eMMC5.1 + 2 GB LP5X
  - 512 MB 266 MHz/16bit PSRAM
- Battery & charging
  - 470 mAh/2.5 V ~ 4.48 V, 2C charging/discharging
  - Wireless charging (CPS4019A)
  - USB2.0 (through debug FPC)
- Sensors:
  - Accelerometer
  - Pressure
  - Ambient light sensor
  - E-compass
  - Body skin temp
  - HRM (PPG+ECG)
- WWAN configurations
  - Global bands, 5G Sub6, APT, UHB
- Audio
  - 3x DMICs
  - 1x smart speaker
  - 1x LRA haptic motor
- Connectivity:
  - Wi-Fi 6 (802.11ax) 1x1
  - UWB Channels 5,9
  - Bluetooth 5.4 + LE
  - NFC+ eSE + eSIM: ST54L
  - GNSS dual band (L1,L5)
  - Satellite NB-NTN (B255, B256, B23)
  - **Physical SIM Card not supported**

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Supported Sensors

Sensors	Part number
Accelerometer/gyro	LSM6DSV32X
Pressure	BMP585
HRM/ECG	MAX86178ENJ+
ALS sensor	OPT3004DNPR
RSB sensor	PAT9401E1
Magnetometer	AK09920C
Body temperature sensor	CT7117(x2)

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WRD6100 Platforms Key chips – SKU2

	SoC	WiFi-BT	RF					
Processor, RFFE, Wi-Fi	SW6100 MEP SKU2	QCC6100	SDR125	QPA6590	QFM6515	QET6105	QAT5625	QAT5655

	Main PMIC	Display PMIC	Wireless charge
PMIC	PW6100	AAT1556	CPS4019A
Memory/NFC	ePOP eMMC+LP5X	PSRAM APS256XXN	NFC ST54L (with eSIM)
Audio	WSA8855 Audio PA	DMICs QMP1000 X 3	

	1.39' Circle Display
DISPLAY	DDIC RM6D030

	ACCE+GYRO	MAGNETO	HRM/ECG	RSB	ALS	PRESSURE	Body temp
Sensors	LSM6DSV32X ST	AK09920C AKM	MAX86178EN J ADI	PAT9401E1 PIXART	OPT3004 TI	BMP585 BOSCH	CT7117 X 2 Sensyslink

QCOM Chip

3rd Chip

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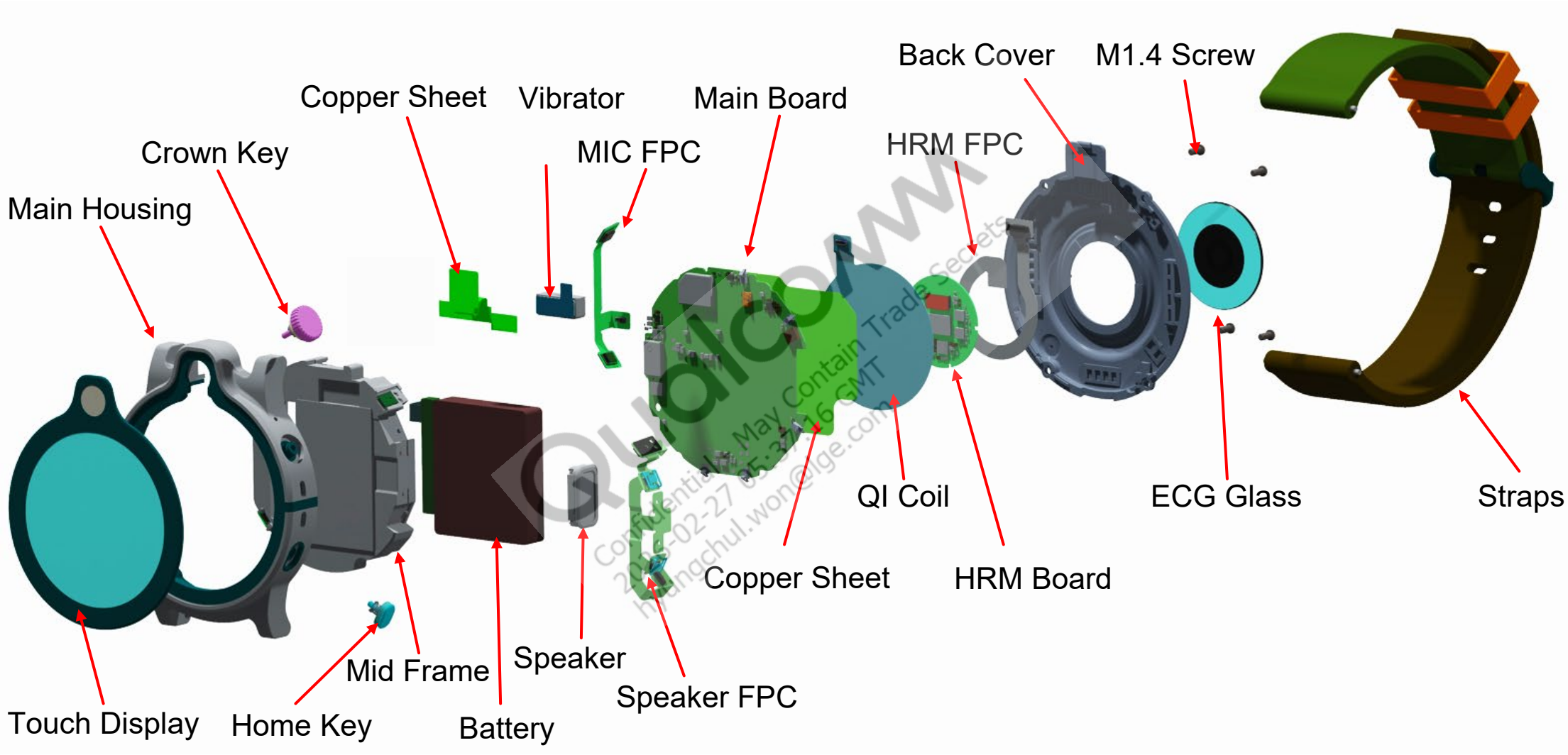
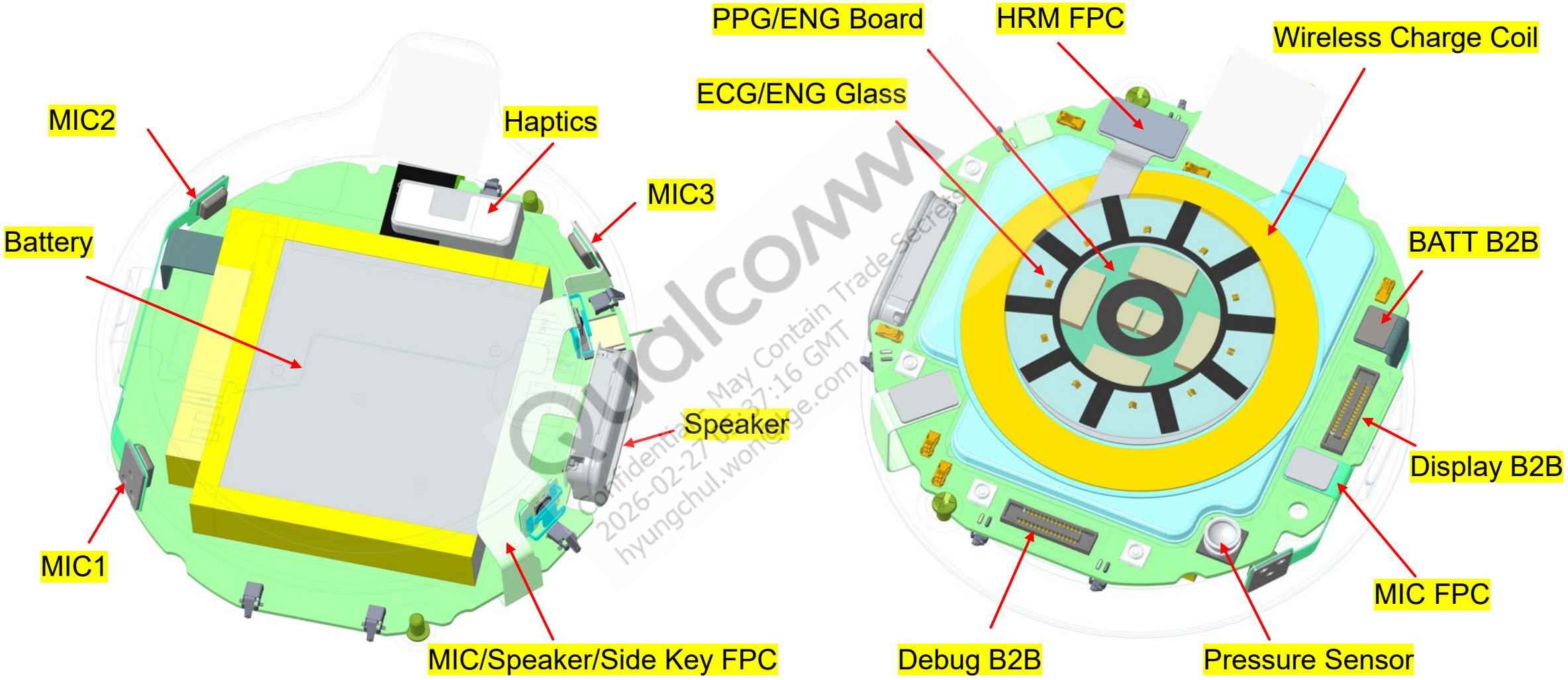
WRD6100 SKU2 Exploded View



WRD6100 SKU2 Mechanical Design – Placement



Top (Display Side)

Bottom

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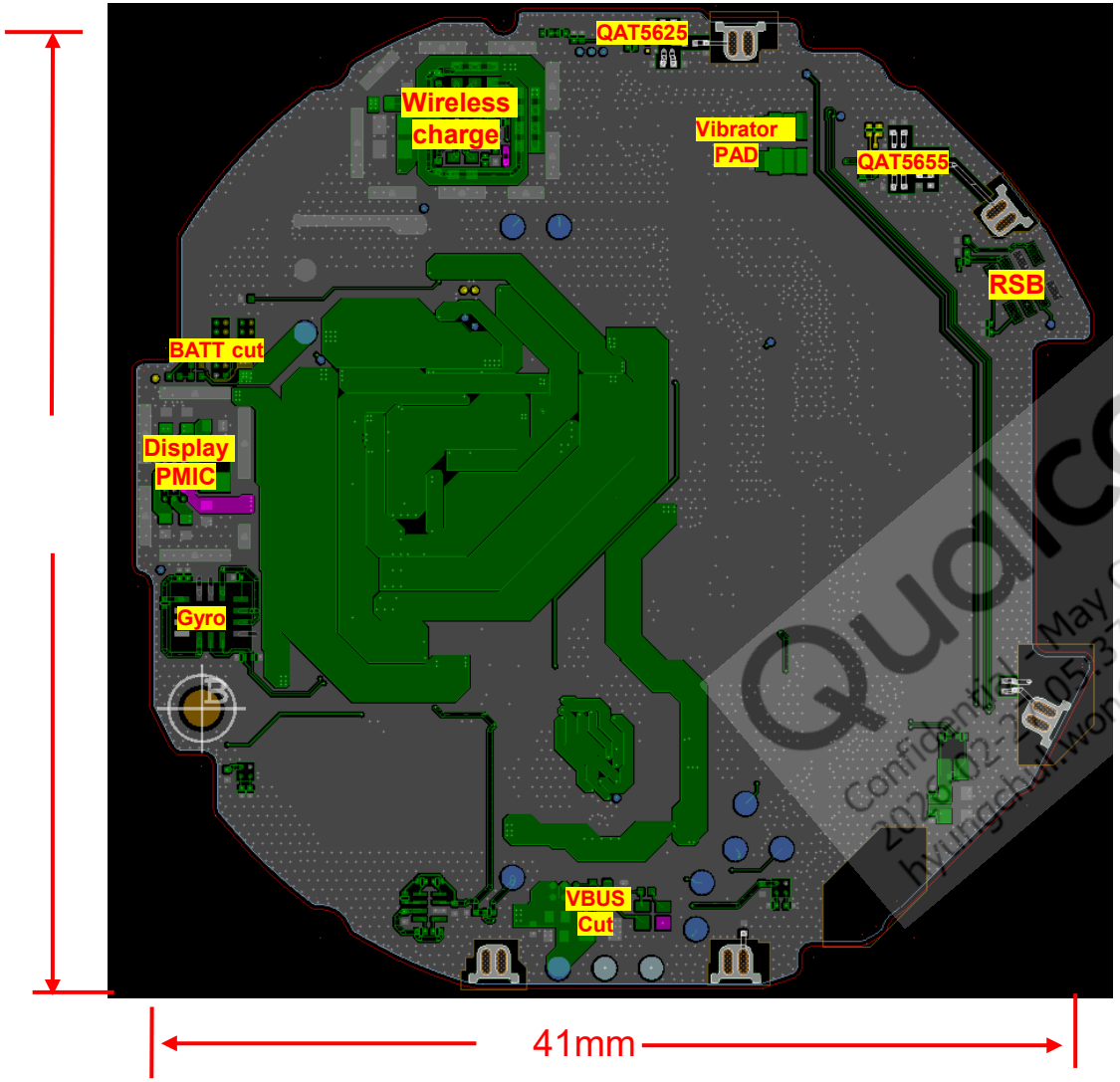
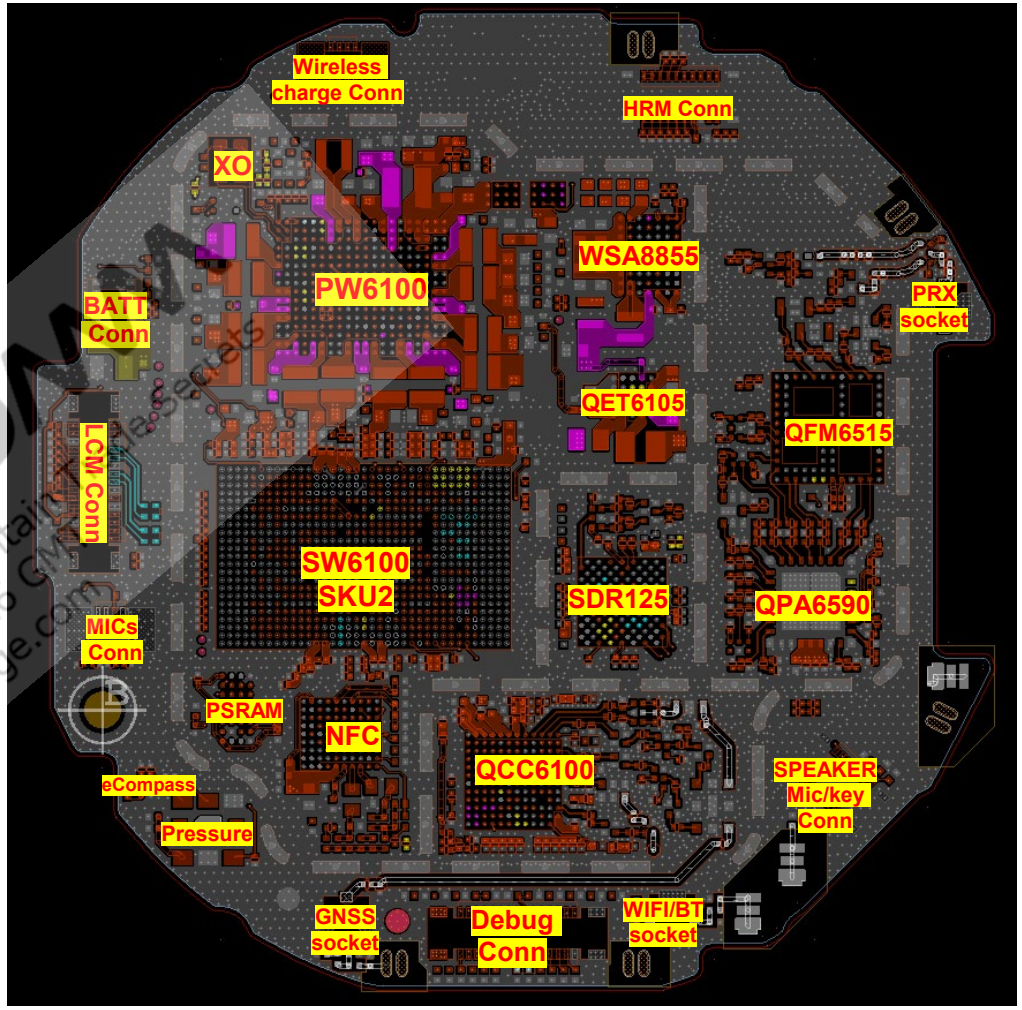
Main Board Overall Dimensions and Placement for SKU2



Top Side (Face To Display)



Bottom Side

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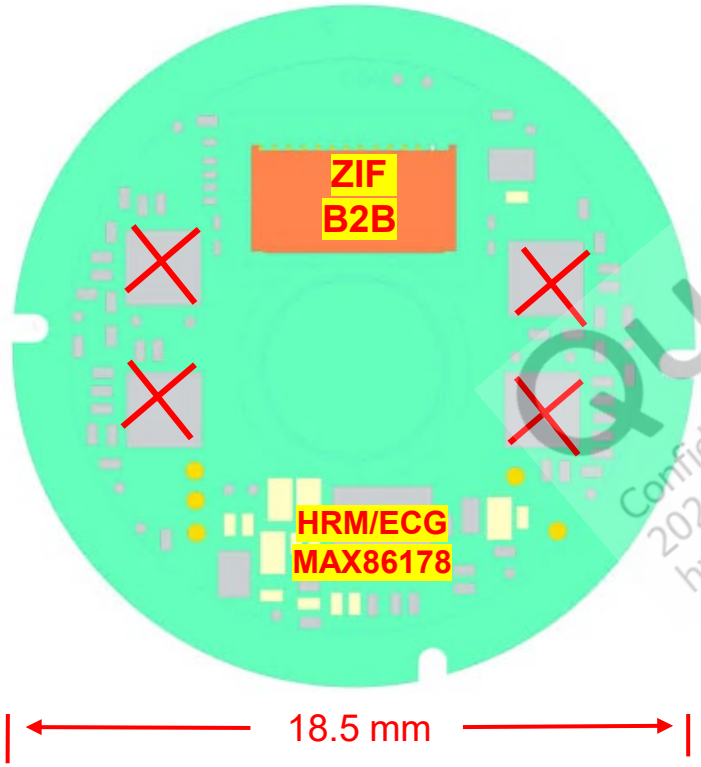
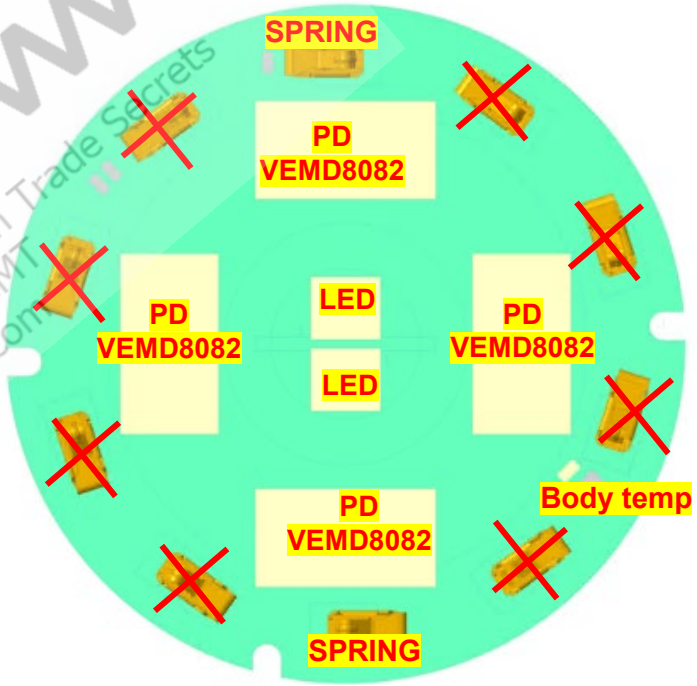
HRM Board Overall Dimensions and Placement SKU2

- 20-79382-H1 board
  - PPG/ECG IC: MAX86178ENJ+
  - 2 x LEDs for transmitting (Green_Red_IR)
  - 4 x PDs for receiving
  - 1 x body temperature sensor



Top Side (Face To Display)



Bottom Side

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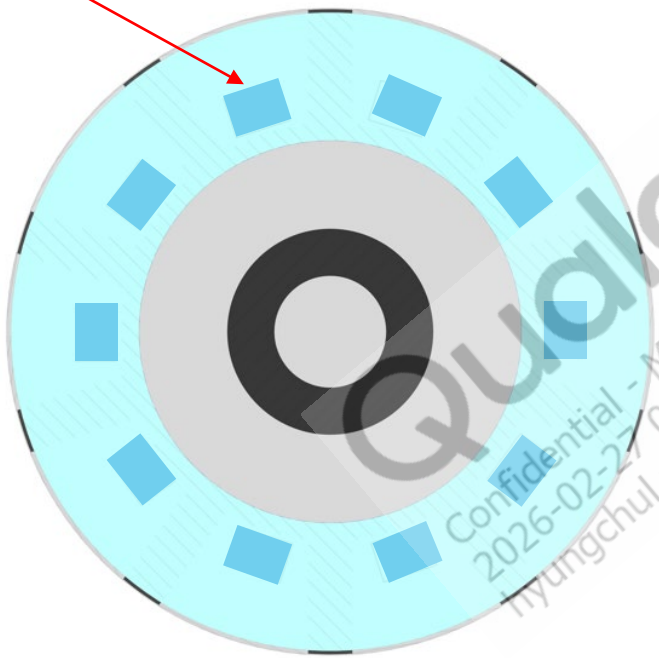
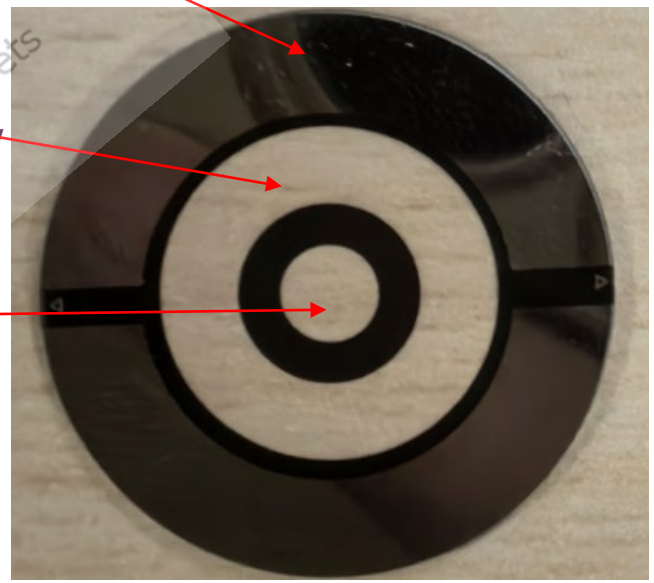
Mechanical PPG/ECG Glass SKU2

- PPG/ECG glass





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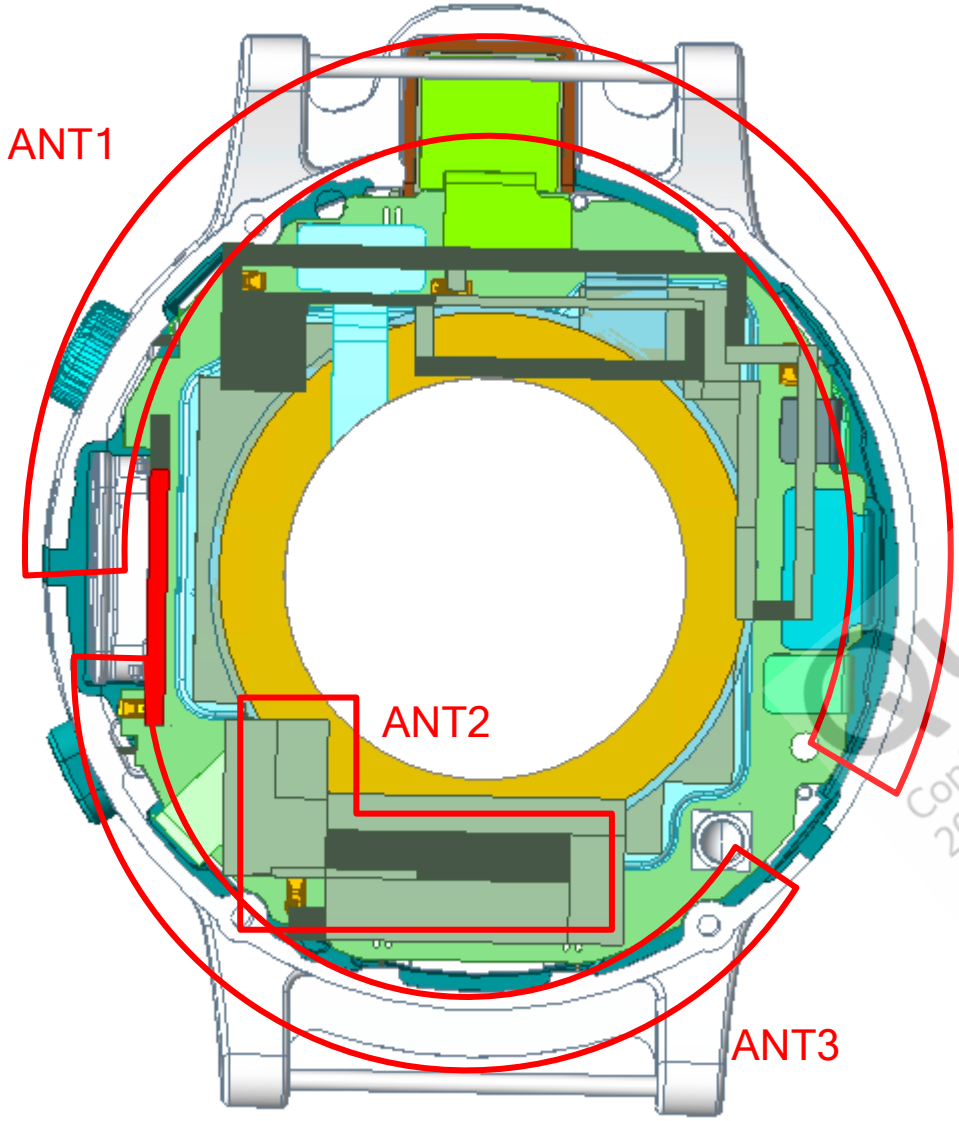
Overall Antenna System SKU2



ANT	Function	Frequency	Antenna Type	Tuner
ANT1	L/M/HB/ Tx0 NTN	703~960 MHz 1525~2700 MHz	Metal ring	QAT5655 QAT5625
ANT2	WIFI (2.4/5/6G) BT	2400~2500 MHz 5100~7125 MHz	FPC	/
ANT3	GNSS L1/L5 UWB	1174 MHz 1559~1606 MHz 6240~8240 MHz	Metal ring	/

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WRD6100 SKU2 Key Component List

Category	Function	Description	Qty	WRD6100 SKU2 MEP 8x13 + PMIC FC Config2	Vendor
Chipset	SoC	Main chip	1	SW6100 MEP 8x13	Qualcomm
	PMIC	Core PMIC	1	PW6100 FC	Qualcomm
Audio PA	Audio PA		1	WSA8855	Qualcomm
Display/camera	AMOLED display	1.39" circle LTPS	1	Module: ATV-D0139MAMBCA6_V0 AUO 1.39" 454*454 DDIC RM6D030 (Raydium)	ANTAVIC
	Touch	InCell Touch	1	Module: ATV-D0139MAMBCA6_V0	ANTAVIC
Sensors	G-sensor + Gyro sensor	6-axis	1	LSM6DSV32X	ST
	Magnetometer	3-axis	1	AK09920C	AKM
	ALS	Ambient light sensor	1	OPT3004DNPR	TI
	Pressure	Pressure sensor	1	BMP585	Bosch
	Body skin temperature	Body skin temperature	1	CT7117AJ4R	Sensyslink
			1	CT7117EJ4R	Sensyslink
	HRM/ECG sensor	Heart rate sensor	1	MAX86178ENJ+	ADI
	RSB sensor	Crown key rotation sensor	1	PAT9401E1	PixArt
	Photodiode	Photodiode	4	VEMD8082	Vishay
	Multi chip LED	Multi chip LED	2	SFH7016	OSRAM
	ENG	ENG(electroneurography)	4	STENG1AX	ST
Memory	ePOP memory	eMMC + LPDDR5X	1	KMZZ60020M-A218	Samsung
	PSRAM	SPI PSRAM	1	APS512XXC-AOB7NI-WB x16 512Mbit 1.2V 266Mhz DDR	AP Memory
Wireless charging	Charging	Wireless charging IC	1	CPS4019A	Convenient POWER
NFC	NFC	NFC	1	ST54L03AEL9CCJ7 (Integrate eSIM)	ST MICROELECTRONICS
AMOLED power IC		AMOLED power IC	1	AAT1556	ADVANCED ANALOG
Clock	Crystal	38.400 MHz	1	EXS00A-CS12755 1.6mmX1.2mm	NDK
	XO	32.768 KHz	1	SIT1566A-JE-0001	Si Time

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WRD6100 SKU2 Key Component List (cont.)

Category	Function	Description	Qty	WRD6100 SKU2 MEP 8x13 + PMIC FC Config2	Vendor
Chipset	SoC	Main chip	1	SW6100 MEP 8x13	Qualcomm
Audio	MIC	Digital MEMS MIC	3	QMP1000	Qualcomm
	Speaker	Speaker	1	FG-1206H2S-6-P	Forgrand
Vibrator	LRA	AC linear resonant actuator	1	BLM0410G-FP249-0RUAJ	Baolong
Battery		Lithium-ion battery	1	AEC492627 Si-Anode/450mAH/2C charger/discharge 2.5V Cut-off	AEC
Cellular	Chipset	RFIC	1	SDR125	
	QAT	Aperture Tuner	1	QAT5625	Qualcomm
		Impedance Tuner	1	QAT5655	
	PA/ASM	4G LMH-PA	1	QPA6590	
		LMH L-FEMID	1	QFM6515	
	QET	QET	1	QET6105	
	LNA	eLNA	1	MXD8921HGW	Maxscend
	Switch	SP2T	1	BGS12WN6E6327XTSA1	Infineon
		SP2T	3	QM12312	Qorvo
	N-Plexer	Diplexer	1	LFD18829MMP1E441	Murata
	Filter	N255/266 TX SAW	1	B7585	RF360
		N255/266 RX SAW	1	B7586	RF360
		LB LPF	1	DEA160960LT-5059A1	TDK
		5G WL Jammer FL	1	DEA162690LT-5064A1	TDK
		BT Notch	1	BJ58A(ES)	RF360
GNSS	Triplexer	GNSS+B255+UWB	1	TPX259236MT-7343A1(ES)	TDK
	GNSS BPF	L1 BPF	2	B39162B7504L210	RF360
		L5 BPF	2	B39122B7505L210	RF360
	GNSS eLNA	L1 eLNA	1	BGA123N6	Infineon
		L5 eLNA	1	BGA125N6	Infineon
WLAN	Chipset	WCN	1	QCC6100	Qualcomm
	Diplexer	Diplexer	1	DPX107250DT-6130B1	TDK
	5G Filter	BPF	1	DEA146138BT-2413B1	TDK
	2.4G Filter	BPF	1	B39242B7530L210	RF360
	UWB Filter	BPF	1	DEA167239BT-2404B1(ES)	TDK
	1P2T Switch	Switch	1	BGS12WN6E6327XTSA1	Infineon

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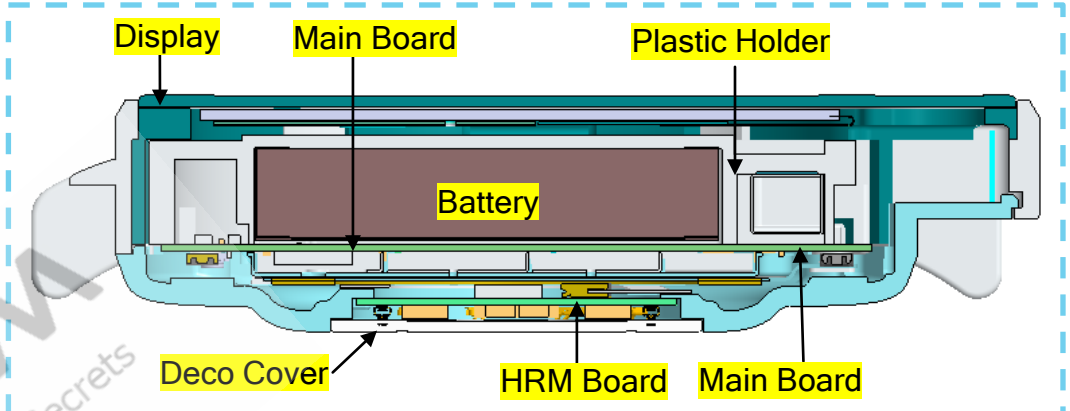
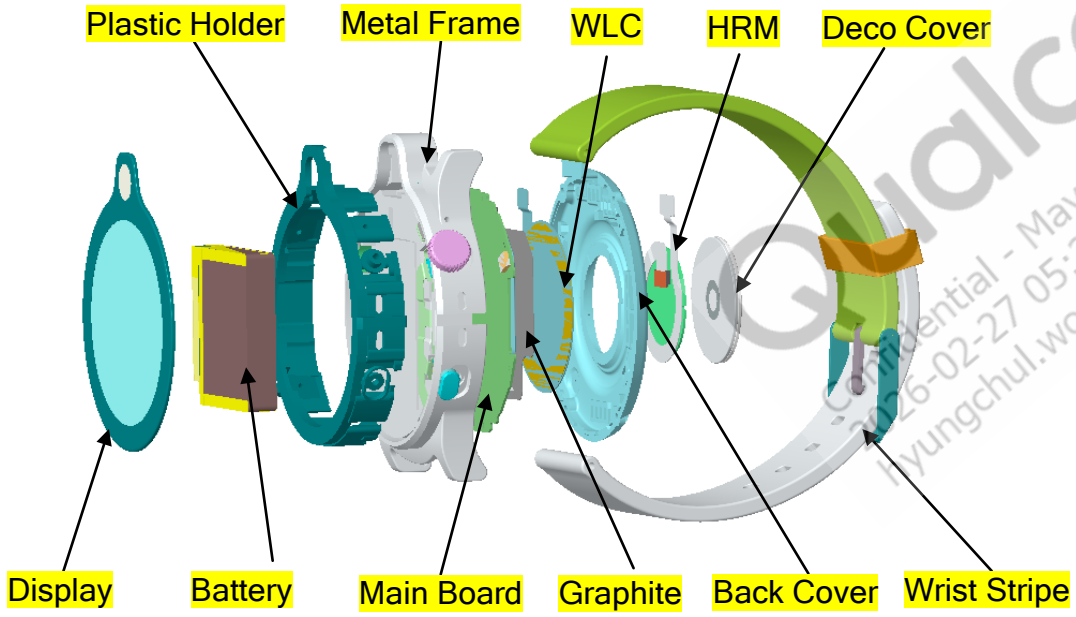
Thermal Solution and Stack-up SKU2

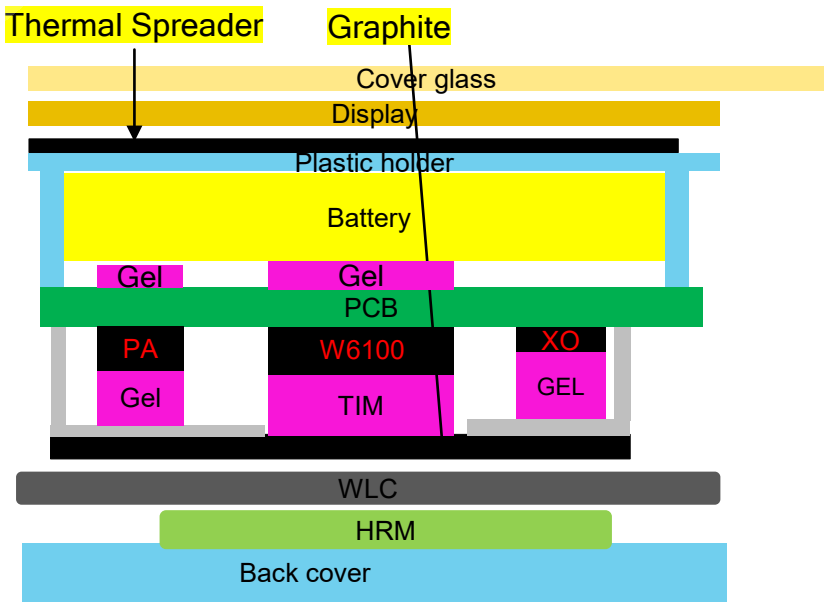
- There is no graphite inside back cover due to antenna clearance.
- Keep gap between main board and battery to prevent battery expansion risk, just add gel opposite area of SW6100 and QPA6590.



Section View



Explode View



Thermal Stack-up

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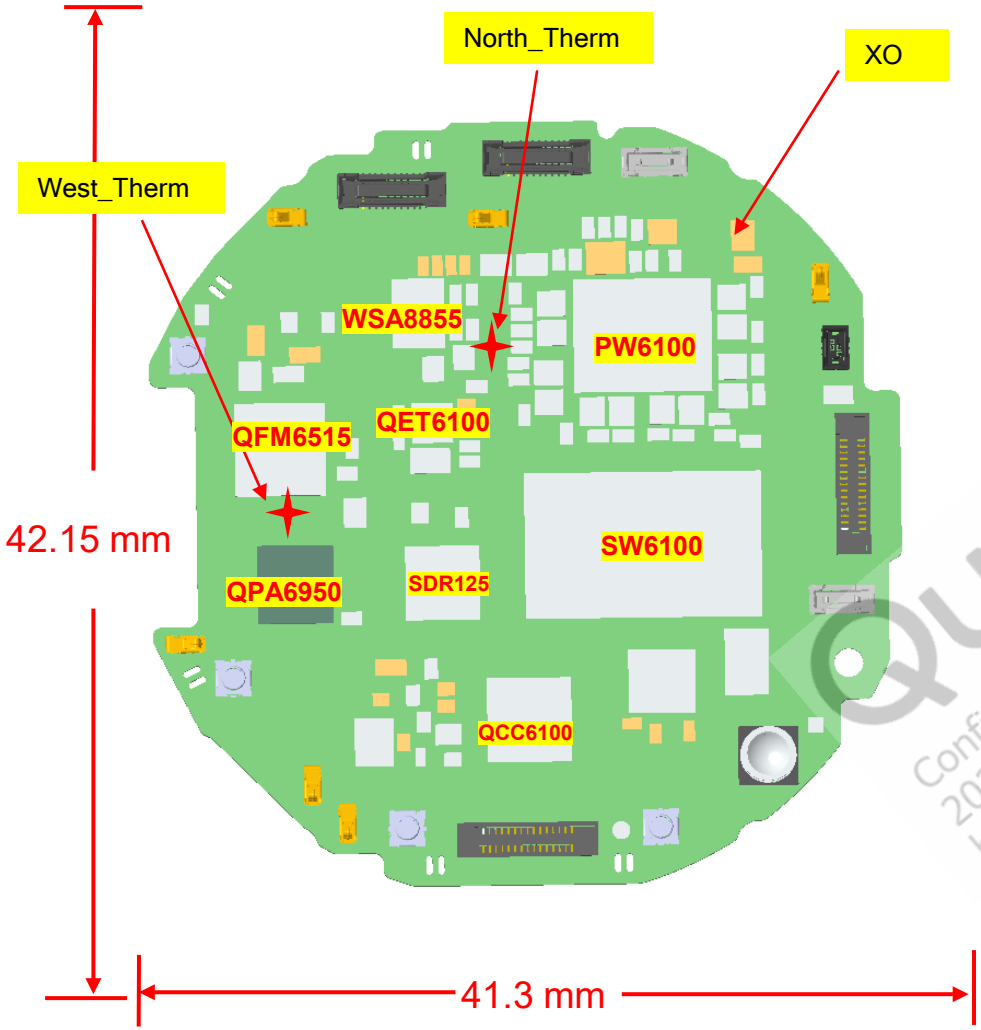
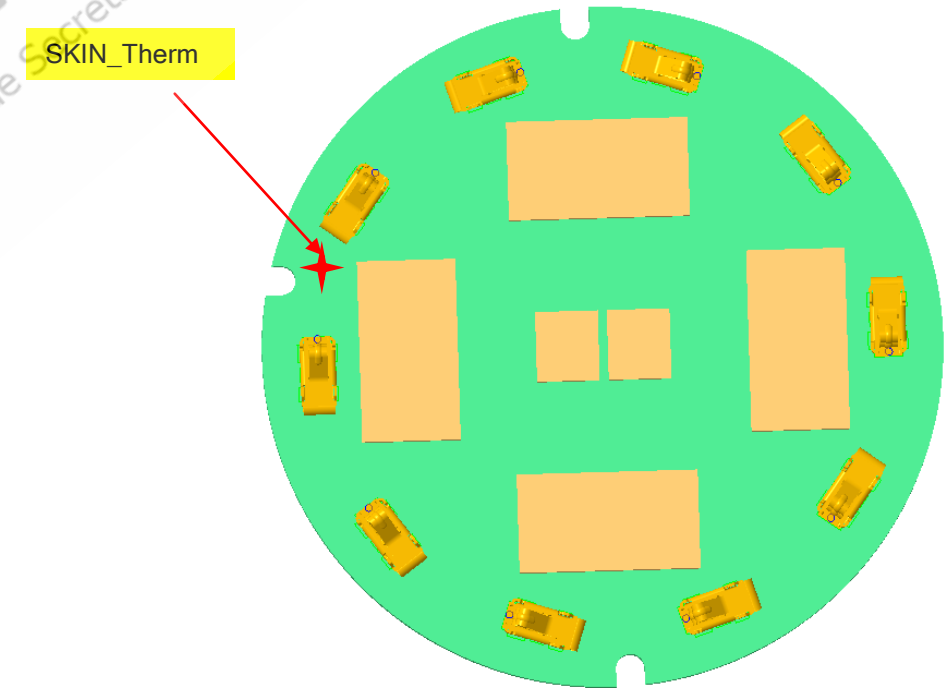
WRD6100 SKU2 Thermistors Configuration



Main Board

PMIC and Pin	Thermistor Function
AMUX_1	NORTH_THERM (GNSS)
AMUX_2	WEST_THERM (RF PA)
AMUX_3	SKIN_THERM (It is on HRM board, close to SKIN)
AMUX_4	Boot option



HRM Board

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Section 6

WRD6100 SKU1/SKU2 Common Features

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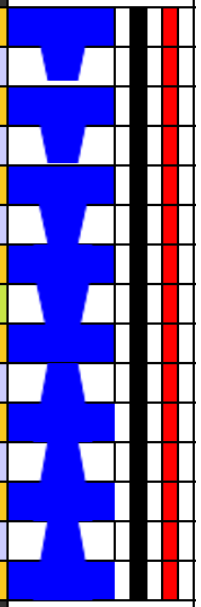
PCB Stack-Up – Main Board (SKU1/SKU2)

- DMICs are connected to MSM through Sound-wire

Layer name	Material type	Finished Thk.	L/S	LVH Pad Size	Tolerance (um)	Er @2G Hz	Experience Er @	Df @ 2G Hz	Copper foil Rz	Etching Factor	Material name
SM Top	Solder ma	20			Min 10	3.7	3.7				PSR-4000D10ME
L1	Copper	20	60/60	70/170	+10/-10				4 ~6um	9um	Copper
	Prepreg	39			+12/-12	3.43	3.03	0.009			EM-39B #1027
L2	Copper	15	40/40	70/170	+7/-7				3 ~4um	7um	Copper
	Prepreg	39			+12/-12	3.43	3.03	0.009			EM-39B #1027
L3	Copper	15	40/40	70/170	+7/-7				3 ~4um	7um	Copper
	Prepreg	39			+12/-12	3.43	3.03	0.009			EM-39B #1027
L4	Copper	15	40/40	70/170	+8/-8				3 ~4um	7um	Copper
	Laminate	51			+6/-6	3.38	2.98	0.01			EM390#1037
L5	Copper	15	40/40	70/170	+8/-8				3 ~4um	7um	Copper
	Prepreg	39			+12/-12	3.43	3.03	0.009			EM-39B #1027
L6	Copper	15	40/40	70/170	+7/-7				3 ~4um	7um	Copper
	Prepreg	39			+12/-12	3.43	3.03	0.009			EM-39B #1027
L7	Copper	15	40/40	70/170	+7/-7				3 ~4um	7um	Copper
	Prepreg	39			+12/-12	3.43	3.03	0.009			EM-39B #1027
L8	Copper	20	60/60	70/170	+10/-10				4 ~6um	9um	Copper
SM Bot	Solder ma	20			Min 10	3.7	3.7				PSR-4000D10ME
		455									



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PCB Design Rules (SKU1/SKU2)

Index	PCB Design Rules	SW6100
A	Line width HDI buildup (outer layers)	60
B	Line space HDI buildup (outer layers)	60
C	uVia pad diameter	170
D	uVia drill diameter	70
E	Mechanical via pad diameter	-
F	Mechanical via drill diameter	-
G	Line width HDI buildup(inner layers)	40
H	Line space HDI buildup(inner layers)	40
I	Soldermask to copper (trace,via,pad,etc.)	37.5
J	Soldermask Dam Width (min web)	50
K	SMD pad to trace (non-SM defined)	90
L	SMD pad to trace (SM defined)	90
SA	PCB Layers	8
SB	**HDI Layer configuration**	**8L full stack**

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DFM Rules for PCB Design (SKU1/SKU2)

Component Body-Body Clearance Limits

Clearance(um)	008004	01005	0201	0402	05035	0603	Other	BGA	Shield
008004	---	---	---	---	---	---	---	---	---
01005	---	150	---	---	---	---	---	---	---
0201	---	150	200	---	---	---	---	---	---
0402	---	250	200	200	---	---	---	---	---
05035	---	250	200	200	200	---	---	---	---
0603	---	250	200	250	250	250	---	---	---
Other	---	250	250	300	300	300	300	---	---
BGA	---	250	250	250	250	250	300	300	---
Shield	---	250	300	300	300	300	300	300	300
Connector	---	300	300	250	250	250	300	300	300
Board Edge	---	300	300	300	300	300	300	300	300

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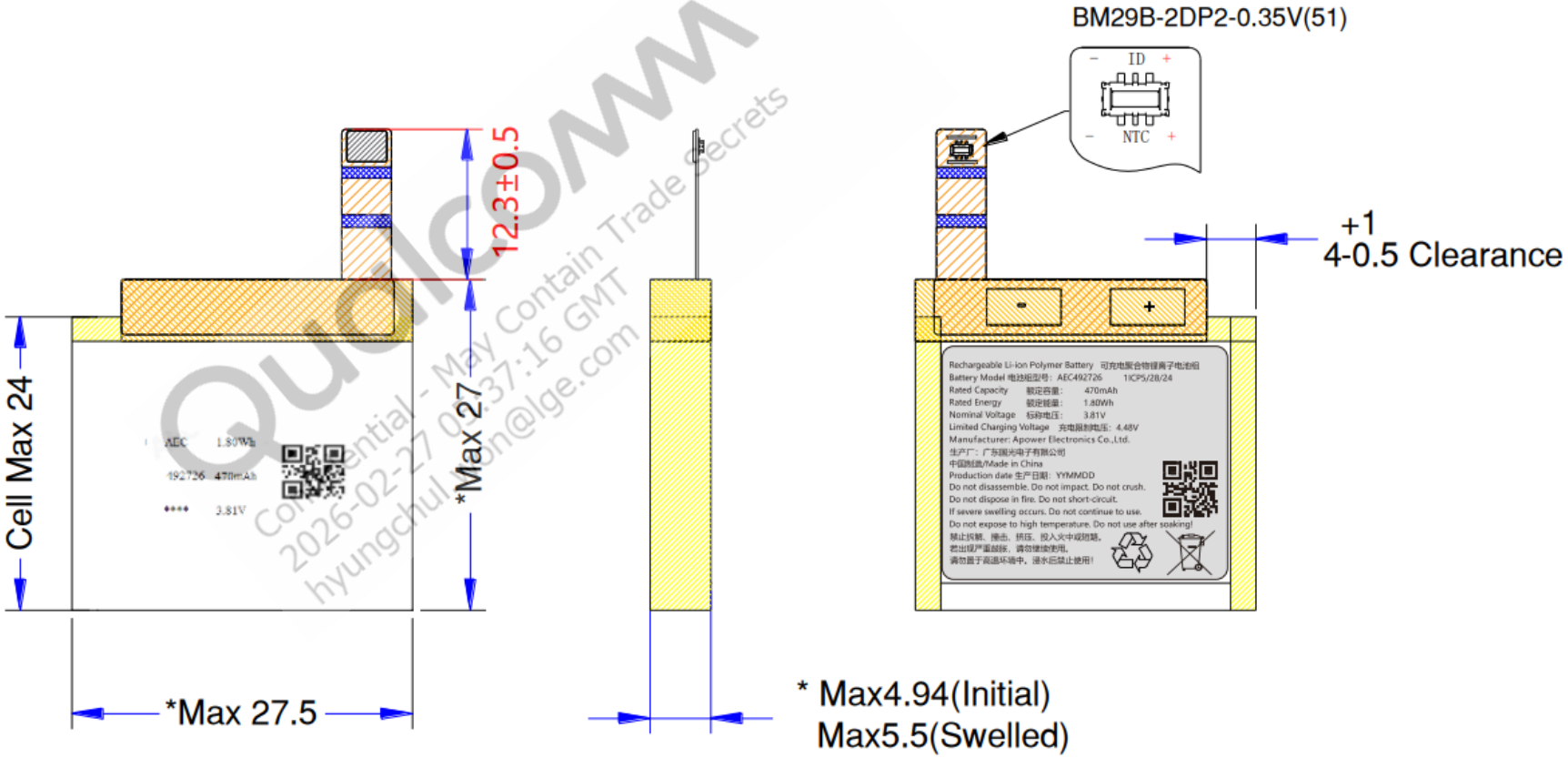
Battery (SKU1/SKU2)

- Part number: AEC492726
- Vendor: Apower Electronics
- 470 mAh 2C fast charging, Li-ion Si-Anode 2.5 V cut off
- Package size Max: 27.0x27
- Connector: 6 Pin BTB



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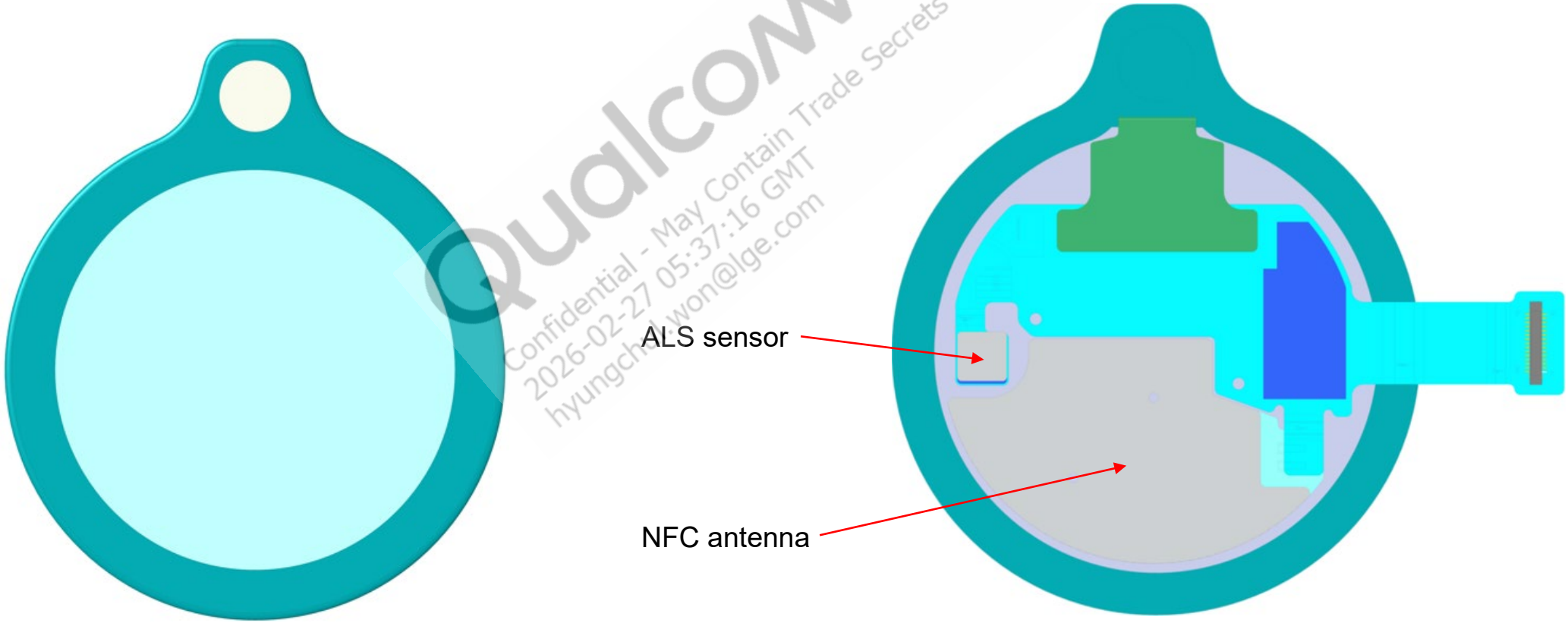
Display (SKU1/SKU2)

- 1.39” LTPS-AMOLED Vendor: ANTAVIC
  - Driver IC: RM6D030
  - Resolution 454x454, 45 fps
  - Connector: 30 pin connector
  - Include NFC antenna and ALS sensor OPT3004



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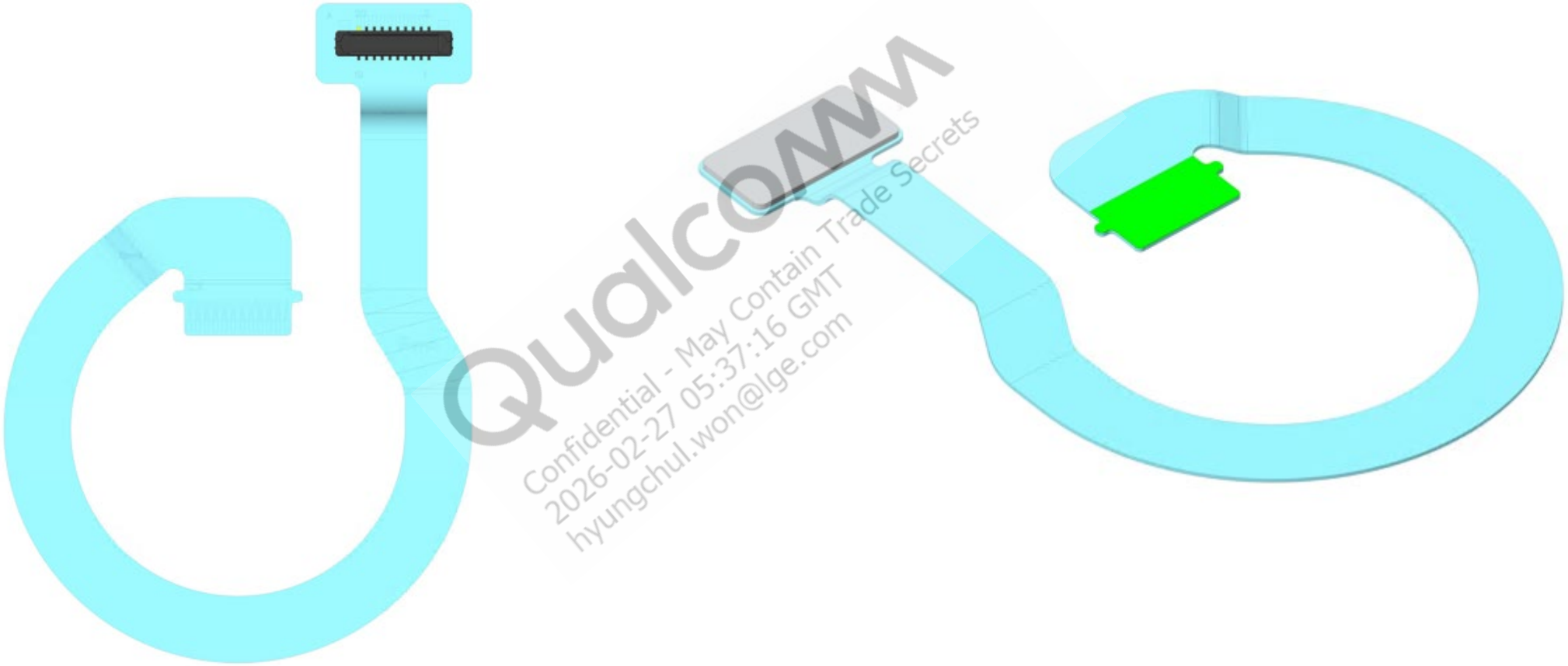
HRM FPC (SKU1/SKU2)

- Connect main board and HRM board



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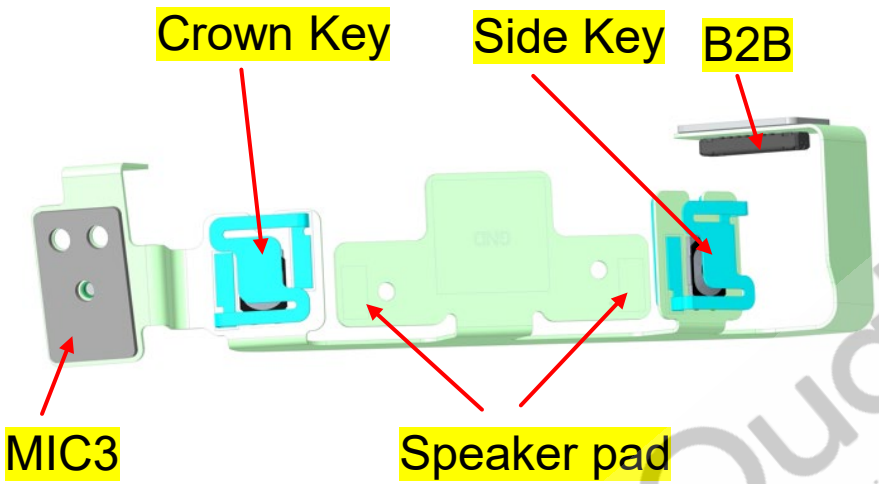
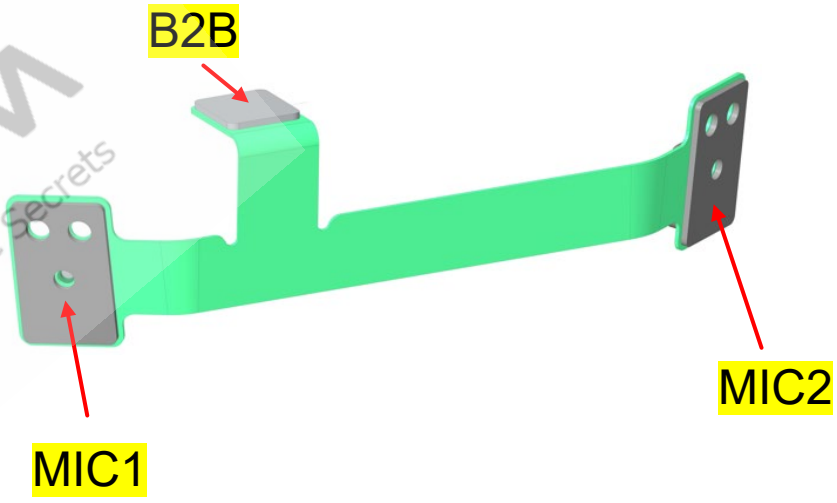
Speaker/ MIC FPC (SKU1/SKU2)

Speaker/MIC/Side Key FPC



- Switch: EVPBB4A9B000
- Vendor: Panasonic
- Connector: 14 Pin BTB

MIC FPC



- Connector: 8 Pin BTB
- QMP1000

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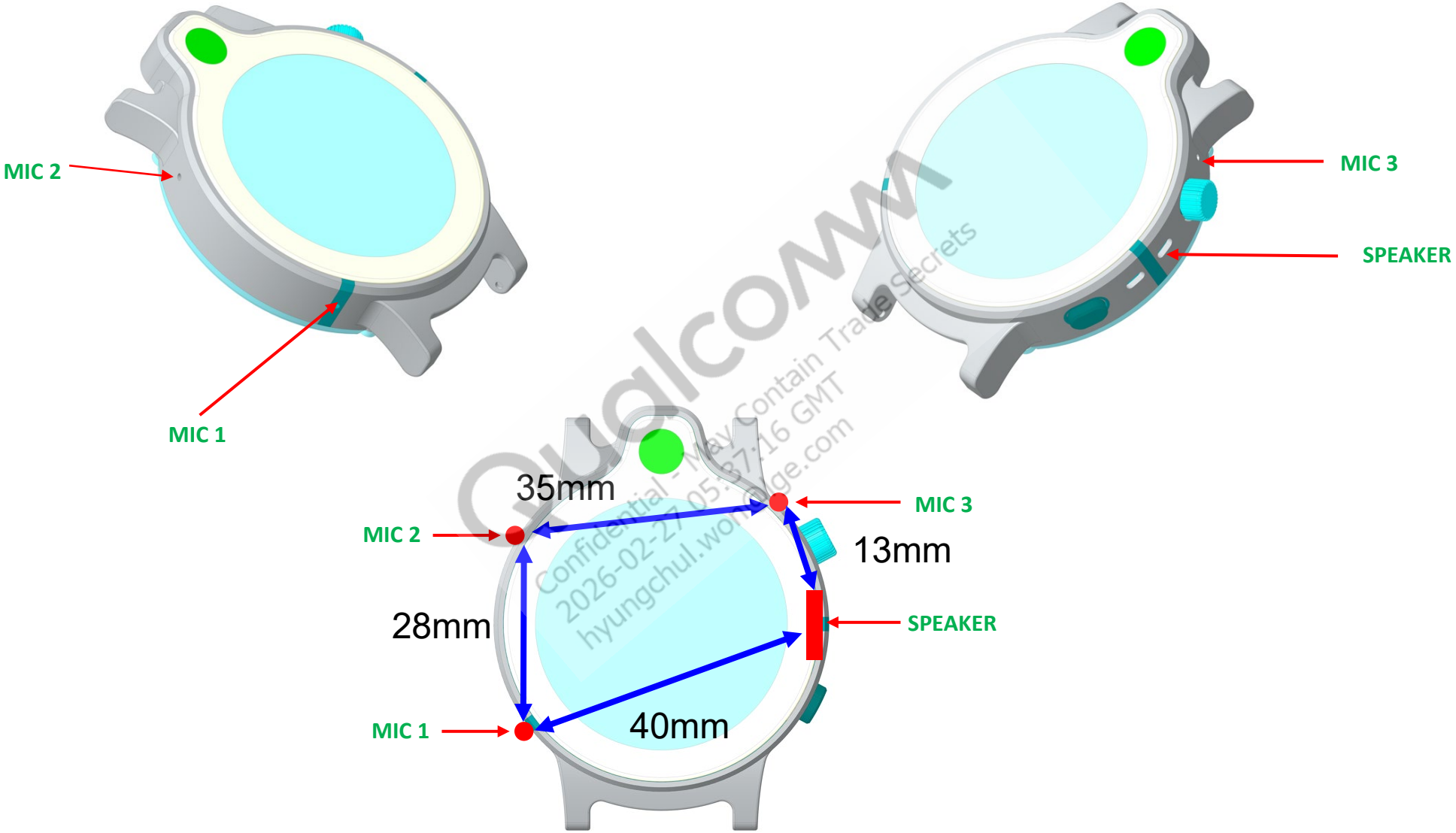
WRD6100 MICs and Speaker location (SKU1/SKU2)



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Section 7

Debug Interface

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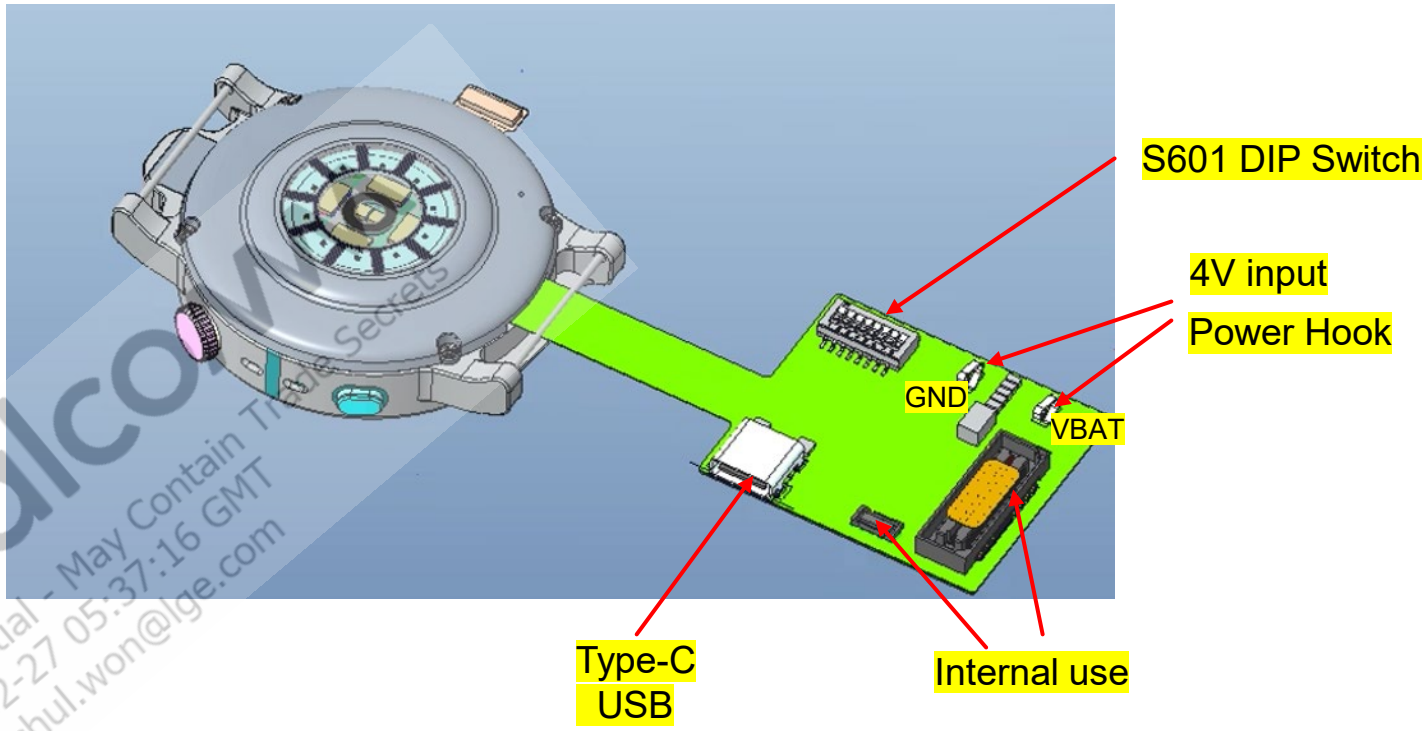
Debug FPC default switch setting

S601 DIP SWITCH CONFIGURATION			
No	**Function**	**Description**	**Default status**
1	FORCE_USB_BOOT	OFF: Floating ON: Pull high (USB download)	OFF
2	BATT_ID	OFF: Disconnected ON: 7.5K(without battery)	OFF
3	BATT_ID	OFF: Disconnected ON: 33K(without battery)	OFF
4	BATT_THERM	OFF: Disconnected ON: 100K	OFF
5	BATT_EN	OFF: Floating (power from Battery) ON: Pull low (power from debug FPC or debug board)	ON
6	RID	OFF: Floating ON: 100K	OFF
7	RID	OFF: Floating ON: 255K	OFF
8	BATT_ID	OFF: Disconnected ON: 7.5K(with battery)	ON



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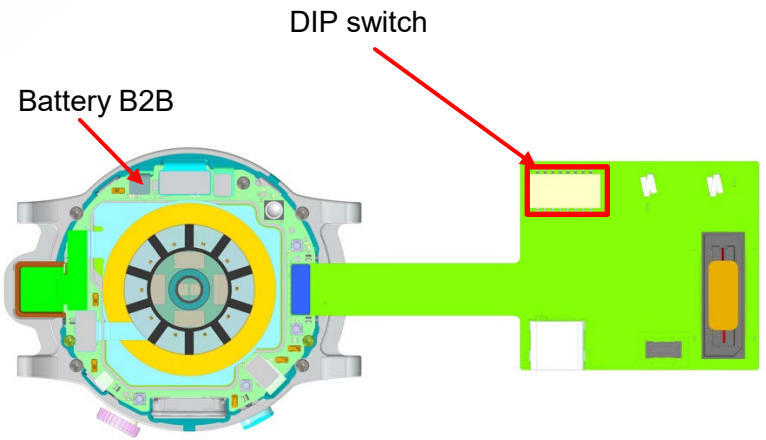
NOTE: Do not connect external power supply when S601-5 is OFF,it will cause external supply charge the battery directly.

Input Power options- Battery

- Disconnect external power supply.
- Set S601-5, S601-8 to OFF status or remove debug FPC.

S601 DIP SWITCH CONFIGURATION			
No	**Function**	**description**	**Default status**
1	FORCE_USB_BOOT	OFF: Floating ON: Pull high (USB download)	OFF
2	BATT_ID	OFF: Disconnected ON: 7.5K(without battery)	OFF
3	BATT_ID	OFF: Disconnected ON: 33K(without battery)	OFF
4	BATT_THERM	OFF: Disconnected ON: 100K	OFF
5	BATT_EN	OFF: Floating (power from Battery) ON: Pull low (power from debug FPC or debug board)	OFF
6	RID	OFF: Floating ON: 100K	OFF
7	RID	OFF: Floating ON: 255K	OFF
8	BATT_ID	OFF: Disconnected ON: 7.5K(with battery)	OFF



NOTE: Do not connect external power supply when S601-5 is OFF, it will cause external supply charge the battery directly.

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